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TABLE 160-2 Antimicrobial Therapy for Pertussis

DRUG	ADULT DAILY DOSE	FREQUENCY	DURATION, DAYS	COMMENTS
Erythromycin estolate	500 mg	3–4 times per day	7–14	Frequent gastrointestinal side effects
Clarithromycin	500 mg	Twice a day	7	—
Azithromycin	500 mg on day 1, 250 mg subsequently	1 daily dose	5	—
Trimethoprim-sulfamethoxazole	160 mg of trimethoprim, 800 mg of sulfamethoxazole	Twice a day	14	For patients allergic to macrolides; data on effectiveness limited

Source: T Tiwari et al: Recommended antimicrobial agents for the treatment and postexposure prophylaxis of pertussis: 2005 CDC guidelines. MMWR Recomm Rep 54(RR-14):1, 2005.

spread by large respiratory droplets. Isolation should continue for 5 days after initiation of macrolide therapy or, in untreated patients, for 3 weeks (i.e., until nasopharyngeal cultures are consistently negative).

PREVENTION

Chemoprophylaxis Because the risk of transmission of *B. pertussis* within households is high, chemoprophylaxis is widely recommended for household contacts of pertussis cases regardless of their immunization status and should be initiated within 21 days of cough onset in the index case. The effectiveness of chemoprophylaxis is supported by several epidemiologic studies of institutional and community outbreaks. In the only randomized, placebo-controlled study, erythromycin estolate (50 mg/kg per day; maximum dose, 1 g/d) was effective in reducing the incidence of bacteriologically confirmed pertussis by 67%; however, there was no decrease in the incidence of clinical disease. Despite these results, authorities continue to recommend chemoprophylaxis, particularly in households with members at high risk of severe disease (children <1 year of age, pregnant women). Data on the use of the newer macrolides for chemoprophylaxis are not available, but these drugs are commonly used because of their increased tolerability and their effectiveness.

Immunization (See also Chap. 123) The mainstay of pertussis prevention is active immunization. Pertussis vaccine became widely used in North America after 1940; the reported number of pertussis cases subsequently fell by >90%. Whole-cell pertussis vaccines are prepared through the heating, chemical inactivation, and purification of whole *B. pertussis* organisms. Despite their efficacy (average estimate, 85%; range for different products, 30–100%), whole-cell pertussis vaccines are associated with adverse events—both common (fever; injection-site pain, erythema, and swelling; irritability) and uncommon (febrile seizures, hypotonic-hyporesponsive episodes). Alleged associations of whole-cell pertussis vaccine with encephalopathy, sudden infant death syndrome, and autism, although not substantiated, spawned an active anti-immunization lobby. The development of acellular pertussis vaccines, which are effective and less reactogenic, has greatly alleviated concerns about the inclusion of pertussis vaccine in the combined infant immunization series.

Although a wide variety of acellular pertussis vaccines were developed, only a few are still marketed widely; all contain pertussis toxoid and filamentous hemagglutinin. One acellular pertussis vaccine also contains pertactin, and another contains pertactin and two types of fimbriae. Adult formulations of acellular pertussis vaccines have been shown to be safe, immunogenic, and efficacious in clinical trials in adolescents and adults and are now recommended for routine immunization of these groups in several countries.

Although whole-cell vaccines are still used extensively in developing regions of the world, acellular pertussis vaccines are used exclusively for childhood immunization in much of the developed world. In light of evidence of early waning of immunity among children who received acellular pertussis vaccine in infancy, the WHO Strategic Advisory Group of Experts (SAGE) recommends that countries using whole-cell pertussis vaccine for the primary infant immunization series continue to do so. In countries using acellular pertussis vaccines in infancy, additional booster immunizations in older children, adolescents, and adults are recommended to prevent pertussis in high-risk infants. Pertussis immunization is also recommended during pregnancy to increase passive transfer of maternal antibodies to the fetus. Studies in high-income countries demonstrate that immunization of women during pregnancy is 90–93% effective at preventing pertussis in infants <2 months of age and is safe. In North America, acellular pertussis vaccines for children are given as a three-dose primary series at 2, 4, and 6 months of age, with a reinforcing dose at 15–18 months of age and a booster dose at 4–6 years of age. Adolescents (11–18 years of age) and all unvaccinated adults should receive a dose of the adult-formulation diphtheria–tetanus–acellular pertussis vaccine. Immunization is specifically recommended for health care providers, individuals in close contact with infants, and women during the third trimester of every pregnancy. Pertussis vaccine coverage among U.S. adolescents was 86.4% in 2015, and coverage among pregnant women was 48.8% in 2015–2016. However, coverage among adults remains low (23.1% in 2015). Further improvements in adult vaccine coverage may permit better control of pertussis across the age spectrum, with collateral protection of infants too young to be immunized. However, more effective vaccines with longer-lasting protection will ultimately be needed to control this disease.

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161 Diseases Caused by Gram-Negative Enteric Bacilli

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GENERAL FEATURES AND PRINCIPLES

The post-antibiotic era has begun. For most people, this is the first time in their lives that an effective treatment for a bacterial infection may not exist. The Enterobacteriaceae are at the forefront of this evolving public health crisis. For example, the Centers for Disease Control and Prevention (CDC) and the World Health Organization (WHO) have designated carbapenem-resistant Enterobacteriaceae (CRE) as representing a threat level of "urgent" and "priority one, critical."

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Enterobacteriaceae are responsible for a significant proportion of the deaths attributed to antimicrobial-resistant bacteria, of which an estimated 23,000 and 25,000 occur annually in the United States and the European Union, respectively, and three to five times as many (per capita) in low- and middle-income countries (e.g., Thailand). These pathogens cause a wide variety of infections involving diverse anatomic sites in both healthy and compromised hosts. Therefore, a thorough knowledge of clinical presentations and appropriate therapeutic choices is necessary for optimal outcomes. *Escherichia coli*, *Klebsiella*, *Proteus*, *Enterobacter*, *Serratia*, *Citrobacter*, *Morganella*, *Providencia*, *Cronobacter*, and *Edwardsiella* are enteric gram-negative bacilli (GNB) within the family Enterobacteriaceae that commonly cause extraintestinal infections. *Salmonella*, *Shigella*, and *Yersinia*, which also are in the family Enterobacteriaceae but more commonly cause gastrointestinal infections, are discussed in **Chaps. 165, 166, and 171**, respectively.

EPIDEMIOLOGY

E. coli, *Klebsiella*, *Proteus*, *Enterobacter*, *Serratia*, *Citrobacter*, *Morganella*, *Providencia*, *Cronobacter*, and *Edwardsiella* are components of the normal animal and human colonic microbiota and/or the microbiota in various environmental habitats, including long-term-care facilities (LTCFs) and hospitals. As a result, except for certain pathotypes of intestinal pathogenic *E. coli*, these genera are global pathogens. The incidence of infection due to these agents is increasing because of the combination of an aging population and increasing antimicrobial resistance. In healthy humans, *E. coli* is the predominant species of GNB in the colonic microbiota, followed by *Klebsiella* and *Proteus*. GNB (primarily *E. coli*, *Klebsiella*, and *Proteus*) can also colonize the oropharynx and intact skin but, in healthy individuals, tend to do so only transiently. By contrast, in LTCFs and hospital settings, a variety of GNB emerge as the dominant colonizers of both mucosal and skin epithelial surfaces, particularly in association with antimicrobial use, severe illness, and extended length of stay. LTCFs are emerging as an important reservoir for resistant GNB. Such colonization with GNB may lead to subsequent extraintestinal infection; for example, oropharyngeal colonization may lead to pneumonia, and colonic/perineal colonization may lead to urinary tract infection (UTI). The use of ampicillin or amoxicillin was associated with an increased risk of subsequent infection due to the hypervirulent pathotype of *Klebsiella pneumoniae* in Taiwan; this association suggests that changes in the quantity or prevalence of colonizing bacteria may significantly influence the risk of infection. *Serratia*, *Enterobacter*, and, less commonly, *Citrobacter* infection may be acquired directly through a variety of infusates (e.g., medications, blood products, non–U.S. Food and Drug Administration [FDA] approved stem cell products). *Edwardsiella* infections are acquired through freshwater and marine environment exposures and are most common in Southeast Asia.

STRUCTURE AND FUNCTION

Enteric GNB possess an extracytoplasmic outer membrane consisting of a lipid bilayer with associated proteins, lipoproteins, and polysaccharides (capsule, lipopolysaccharide). The outer membrane interfaces with the external environment, including the human host. A variety of components of the outer membrane are critical determinants in pathogenesis (e.g., capsule) and antimicrobial resistance (e.g., permeability barrier, efflux pumps). In addition, secreted products play an important role in both host infection (e.g., iron acquisition molecules) and environmental niche survival and colonization (e.g., type VI secretion systems).

PATHOGENESIS

Multiple bacterial virulence factors are required for the pathogenesis of infections caused by GNB. Possession of specialized virulence genes defines pathogens and enables them to infect the host efficiently. Hosts and their cognate pathogens have been co-adapting throughout evolutionary history. During the host–pathogen "chess match" over time, various and redundant strategies have emerged in both the pathogens and their hosts **(Table 161-1)**.

TABLE 161-1 Interactions of Extraintestinal Pathogenic *Escherichia coli* with the Human Host: A Paradigm for Extracellular, Extraintestinal Gram-Negative Bacterial Pathogens

BACTERIAL GOAL	HOST OBSTACLE	BACTERIAL SOLUTION
Extraintestinal attachment	Flow of urine, mucociliary escalator	Multiple adhesins (e.g., type 1, S, and F1C fimbriae; P pili)
Nutrient acquisition for growth	Nutrient sequestration (e.g., iron via intracellular storage and extracellular scavenging via lactoferrin and transferrin)	Cellular lysis (e.g., hemolysin), multiple mechanisms for competing for iron (e.g., siderophores) and other nutrients
Initial avoidance of host bactericidal activity	Complement, phagocytic cells, antimicrobial peptides	Capsular polysaccharide, lipopolysaccharide
Dissemination (within host and between hosts)	Intact tissue barriers	Irritant tissue damage resulting in increased excretion (e.g., toxins such as hemolysin), invasion of brain endothelium
Late avoidance of host bactericidal activity	Acquired immunity (e.g., specific antibodies), treatment with antibiotics	Cell entry, acquisition of antimicrobial resistance

Intestinal pathogenic (diarrheagenic) mechanisms are discussed below. The members of the Enterobacteriaceae family that cause extraintestinal infections are primarily extracellular pathogens and therefore share certain pathogenic features. The two principal components of host defense against Enterobacteriaceae, regardless of species, are innate immunity (including intact skin and mucosal barriers; the withholding of nutrients; and the activities of complement, antimicrobial peptides, and professional phagocytes) and humoral immunity. Both susceptibility to and severity of infection are increased with dysfunction or deficiencies of these host components. By contrast, the virulence traits of intestinal pathogenic *E. coli*—i.e., the distinctive strains that can cause diarrheal disease—are for the most part different from those of extraintestinal pathogenic *E. coli* (ExPEC) and other GNB that cause extraintestinal infections. This distinction reflects site-specific differences in host environments, defense mechanisms, and physiological derangements that lead to disease.



A given enterobacterial strain usually possesses multiple adhesins for binding to a variety of host cells (e.g., in *E. coli*: type 1, S, and F1C fimbriae; P pili). Nutrient acquisition (e.g., of iron via siderophores) requires many genes that are necessary but not sufficient for pathogenesis. The ability to resist the bactericidal activity of complement and phagocytes in the absence of antibody (e.g., as conferred by capsule or the O antigen component of lipopolysaccharide) is one of the defining traits of an extracellular pathogen. Tissue damage (e.g., as mediated by *E. coli* hemolysin) may facilitate nutrient acquisition and spread within the host. Without doubt, many important virulence genes await identification.

The ability to induce septic shock is another defining feature of these genera. GNB are the most common causes of this potentially lethal syndrome. Pathogen-associated molecular pattern molecules (PAMPs; e.g., the lipid A moiety of lipopolysaccharide) stimulate a proinflammatory host response via pattern recognition receptors (e.g., Toll-like or C-type lectin receptors) that activate host defense signaling pathways; if overly exuberant, this response results in shock **(Chap. 304)**. Direct bacterial damage of host tissue (e.g., by toxins) or collateral damage from the host response can result in the release of damage-associated molecular pattern molecules (DAMPs; e.g., HMGB1) that can propagate a detrimental proinflammatory host response.

Many antigenic variants (serotypes) exist in most genera of GNB. For example, *E. coli* has >150 O (somatic) antigens, 80 K (capsular) antigens, and 53 H (flagellar) antigens. This antigenic variability, which permits immune evasion and allows recurrent infection by different strains of the same species, has impeded vaccine development **(Chap. 123)**.

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INFECTIOUS SYNDROMES

Depending on both the host and the pathogen, GNB can infect nearly every organ or body cavity. *E. coli* can cause either intestinal or extraintestinal infection, depending on the particular pathotype, and *Edwardsiella tarda* can cause both intestinal and extraintestinal infection. *Klebsiella* causes primarily extraintestinal infection, but a toxin-producing variant of *Klebsiella oxytoca* has been associated with hemorrhagic colitis, and *Providencia alcalifaciens* and *Escherichia albertii* have been associated with gastroenteritis.

E. coli and—to a lesser degree—*Klebsiella* account for most extraintestinal infections due to GNB. These species (for *K. pneumoniae*, primarily its hypervirulent pathotype) are the most virulent pathogens within this group, as demonstrated by their ability to cause severe infections in healthy, ambulatory hosts from the community. However, the other genera of GNB are also important extraintestinal pathogens, especially among LTCF residents and hospitalized patients, in large part because of the intrinsic or acquired antimicrobial resistance of these organisms and the increasing number of individuals with compromised host defenses. The mortality rate is substantial in many GNB infections and correlates with severity of illness, underlying host status, and in some cases the antimicrobial resistance of the infecting pathogen, which can result in suboptimal therapy. Especially problematic are pneumonia, sepsis, and septic shock (arising from any site of infection), for which the associated mortality rates are 20–60%.

DIAGNOSIS

Isolation of GNB from sterile sites almost always implies infection, whereas their isolation from nonsterile sites, particularly open wounds and the respiratory tract, requires clinical correlation to differentiate colonization from infection. Clinical microbiology laboratories are increasingly incorporating newer diagnostic methodologies (e.g., matrix-assisted laser desorption–ionization–time-of-flight mass spectrometry [MALDI-TOF-MS] and polymerase chain reaction [PCR]) and immunoassays to enhance the sensitivity, accuracy, and rapidity of reporting on pathogen identification and resistance genes. This information can be used to increase the timeliness of initiation and/or the accurate selection of empirical antimicrobial therapy, thereby improving outcomes.

TREATMENT

Principles Guiding Treatment in the Era of Increasing Antimicrobial Resistance

(See also Chap. 144) Initiation of appropriate empirical antimicrobial therapy early in the course of infections due to GNB (particularly the more serious ones) leads to improved outcomes. The ever-increasing prevalence of multidrug-resistant (MDR) and extensively drug-resistant (XDR) GNB; the lag between published and current resistance rates; and variations in antimicrobial susceptibility by species, geographic location, regional antimicrobial use, and hospital site (e.g., intensive care units [ICUs] vs wards) necessitate familiarity with evolving patterns of antimicrobial resistance for the selection of appropriate empirical therapy.

Patient factors predictive of resistance in a given isolate include recent antimicrobial use, a health care association (e.g., recent or ongoing hospitalization, dialysis, residence in an LTCF, transplant, hematologic malignancy), or international travel (e.g., to Asia, Latin America, Africa, Eastern Europe). Resistance rates will almost certainly increase over time and will likely be higher than shown here by the time this chapter is published. Of concern are an increasing number of reports on resistant Enterobacteriaceae causing infections in ambulatory patients without known risk factors.

In this era of increasing antimicrobial resistance, it is critical to culture the primary site of infection before initiating antimicrobial therapy and, for systemically ill patients, to obtain blood cultures. In vitro testing may not always detect antimicrobial resistance; therefore, it is important to assess the patient's clinical response to treatment. Moreover (see discussion of AmpC β-lactamases below), resistance may emerge during therapy. In addition, drainage of abscesses, resection of necrotic tissue, and removal of infected foreign bodies, sometimes referred to collectively as "source control," are often required for cure.

For appropriately selected patients, it may be prudent initially, pending antimicrobial susceptibility results, to use two potentially active agents as a way to increase the likelihood that at least one agent will be active against the patient's organism. If broad-spectrum treatment has been initiated, it is important to switch to the most appropriate narrower-spectrum agent once antimicrobial susceptibility results become available. Such responsible antimicrobial stewardship should help disrupt the ever-escalating cycle of selection for increasingly resistant bacteria, plus decrease the likelihood of *Clostridioides difficile* infection, decrease costs, and maximize the useful longevity of available antimicrobial agents. Likewise, it is important to avoid treatment of patients who are colonized but not infected (e.g., who have a positive sputum culture without evidence of pneumonia, or a positive urine culture without clinical manifestations of UTI).

At present, the most reliably and broadly active antimicrobial agents in vitro against Enterobacteriaceae are the carbapenems (excepting imipenem, to which the Proteeae [*Proteus*, *Morganella*, *Providencia*] are intrinsically resistant); the aminoglycosides amikacin and plazomicin (excepting the Proteeae); the fourth-generation cephalosporin cefepime; the β-lactamase inhibitor combination agents piperacillin-tazobactam, ceftolozane-tazobactam, ceftazidime-avibactam, meropenem-vaborbactam; and the novel cephalosporin-siderophore cefiderocol. A limitation of imipenem/cilastatin-relebactam; the tetracycline derivatives tigecycline, omadacycline, and eravacycline; and the polymyxins B and E (colistin) (which are otherwise very active) is their poor activity against the Proteeae and *Serratia*. Furthermore, the tetracycline derivatives achieve suboptimal concentrations at several anatomic sites (including urine and blood). Clinical data are limited for cefiderocol outside of UTIs; thus, caution is in order for serious infections.

The number of antimicrobial agents effective against certain Enterobacteriaceae is shrinking, and truly pan-resistant GNB exist. Accordingly, the currently available antimicrobial drugs must be used judiciously. Extensive resistance to available agents may leave the clinician with few or no ideal therapeutic options. However, use of a regimen that takes into account the site of infection, achievable drug levels at that site (e.g., higher concentrations of many agents in urine), and pharmacodynamically guided administration strategies (e.g., prolonged infusion of β-lactam agents to maintain drug levels above the minimal inhibitory concentration [MIC]) may increase the chance for a successful outcome. In the near future, point-of-care identification of resistance mechanisms in GNB will enable a strain-specific, patient-specific precision medicine–based treatment approach that would be predicted to improve outcome.

GNB are commonly involved in polymicrobial infections, in which the role of each individual pathogen is uncertain **(Chap. 177)**. Although some GNB are more pathogenic than others, it is usually prudent, if possible, to design an antimicrobial regimen active against all of the GNB identified, because each is typically capable of pathogenicity in its own right. For patients treated initially with a broad-spectrum empirical regimen, the regimen should be de-escalated as expeditiously as possible once susceptibility results are known and the patient has responded to therapy.

Treatment duration is best individualized based on underlying host status and site of infection. However, for selected non–critically ill patients with source control and a satisfactory clinical response to therapy, 7 days of treatment may suffice.

ANTIMICROBIAL TREATMENT AND RESISTANCE MECHANISMS

The most common resistance mechanisms possessed by Enterobacteriaceae are summarized in **Table 161-2**. However, enzymatic hydrolysis (e.g., β-lactamases, of which >3000 variants have been described) and modification of antimicrobials are the major

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TABLE 161-2 Common Antimicrobial Resistance Mechanisms Possessed by the Enterobacteriaceae

MECHANISM	ANTIMICROBIALS MOST SIGNIFICANTLY AFFECTED	COMMON MEDIATORS OF RESISTANCE
Efflux	Tetracyclines, fluoroquinolones (FQ)	Efflux pumps
Decreased permeability	Fosfomycin	Alterations in uptake system
Target site alteration or over-production	FQ, trimethoprim-sulfamethoxazole (TMP-SMX), and polymyxins	DNA gyrase or topoisomerase IV for FQ; enzymes for folic acid synthesis for TMP-SMX Lipid A for polymyxins
Enzymatic hydrolysis of antimicrobials	Penicillins, cephalosporins, cephamycins, carbapenems	Broad-spectrum β-lactamases (e.g., TEM, SHV) ESBLs (e.g., CTX-M, modified TEM and SHV) AmpC β-lactamases Carbapenemases (e.g., serine based KPC, SME, OXA; and metallo-based NDM, VIM, IMP)
Enzymatic modification of antimicrobials	Aminoglycosides	AAC, ANT, APH

Abbreviations: AAC, *N*-acetyltransferases; ANT, O-adenyltranferases; APH, *O*-phosphotransferases; CTX, cefotaxime β-lactamase; ESBL, extended-spectrum β-lactamase; IMP, active on imipenem; KPC, *Klebsiella pneumoniae* carbapenemase; NDM, New Delhi metallo-β-lactamase; OXA, oxacillinase; SHV, sulfhydryl reagent variable β-lactamase; SME, *Serratia marcescens* enzyme; TEM, temoniera β-lactamase; VIM, Verona integron-mediated metallo-β-lactamase.

mediators of resistance in GNB and will be discussed in detail below. Importantly, it is becoming increasingly recognized that MDR and XDR GNB often possess multiple plasmids and genes that encode for multiple β-lactamases.

Broad-spectrum β-lactamases mediate resistance to many penicillins and first-generation cephalosporins and are frequently expressed in enteric GNB. These enzymes are inhibited by β-lactamase inhibitors (e.g., clavulanate, sulbactam, tazobactam, avibactam). In their wild-type form, they do not hydrolyze third- and fourth-generation cephalosporins or cephamycins (e.g., cefoxitin).

Extended spectrum β-lactamases (ESBLs) are modified broad-spectrum enzymes that hydrolyze third-generation cephalosporins, aztreonam, and (in some instances) fourth-generation cephalosporins, in addition to the drugs hydrolyzed by broad-spectrum β-lactamases. GNB that express ESBLs may also exhibit porin mutations that result in decreased uptake of relevant β-lactam agents (cephalosporins, β-lactam/β-lactamase inhibitor combinations, and carbapenems), further reducing susceptibility to these agents. The prevalence of acquired ESBL production, particularly of CTX-M-type enzymes, is increasing in GNB worldwide, largely due to the presence of the corresponding genes on transferable (conjugal) plasmids, which also variably confer or are associated with resistance to fluoroquinolones, trimethoprim-sulfamethoxazole (TMP-SMX), aminoglycosides, tetracyclines, and (more recently) fosfomycin. To date, ESBLs are most prevalent in *E. coli* (especially ST131), *K. pneumoniae*, and *K. oxytoca*, but these enzymes can occur in all Enterobacteriaceae. The approximate regional prevalence of ESBL-producing GNB currently follows a descending gradient as follows: China > Eastern Europe > other parts of Asia (e.g., India) > Latin America and Africa > Western Europe, the United States, Canada, and Australia. Travel to high-prevalence regions increases the likelihood of colonization with these strains. The incidence of community-acquired infections due to ESBL-producing *Enterobacteriaceae* has increased worldwide, including in the United States.

Carbapenems are the most reliably active β-lactam agents against ESBL-expressing strains. Piperacillin-tazobactam, when active in vitro, has been used as a carbapenem-sparing alternative, but recent data from the MERINO trial do not support its use for bloodstream infections. Ceftazidime-avibactam, ceftolozane-tazobactam (less active against *Klebsiella*, *Enterobacter*, and *Citrobacter*), meropenem-vaborbactam, imipenem/cilastatin-relebactam, and plazomicin are active against most ESBL-producing strains and have limited clinical data that support potential utility. The roles for tigecycline, eravacycline, and omadacycline are unclear despite these agents' excellent in vitro activity against most Enterobacteriaceae; however, they are inactive against *Proteus*, *Morganella*, *Providencia*, and *Serratia*.

Oral options for the treatment of ESBL-expressing strains are limited. Fosfomycin, nitrofurantoin (for *E. coli*, 75–90% susceptible), pivmecillinam (not available in the United States), and omadacycline are the most reliably active agents. Older tetracyclines (e.g., doxycycline and minocycline) are also often active, although urine levels may be insufficient and clinical experience with gram-negative infections is limited.

AmpC β-lactamases, when induced or stably derepressed to high levels of expression, confer resistance to the same substrates as do ESBLs, plus to the cephamycins (e.g., cefoxitin and cefotetan). The genes encoding these enzymes are primarily chromosomal and therefore may not exhibit the linked resistance to TMP-SMX, aminoglycosides, and tetracyclines that is common with ESBLs. These enzymes are problematic for the clinician: resistance may develop during therapy with third-generation cephalosporins and result in clinical failure, particularly in the setting of bacteremia. Although chromosomal AmpC β-lactamases are present in nearly all members of the Enterobacteriaceae family (with the notable exceptions of *K. pneumoniae*, *K. oxytoca*, and *Proteus mirabilis*), the risk of clinically significant induction of high-level expression or selection of stably derepressed mutants with cephalosporin treatment is not uniform across species, being greatest with *Enterobacter cloacae*, *Klebsiella* (formerly *Enterobacter*) *aerogenes*, *Citrobacter freundii*, and *Hafnia alvei*, and less with *Serratia marcescens*, *Providencia*, and *Morganella morganii*. In addition, rare strains of *E. coli*, *K. pneumoniae*, and other Enterobacteriaceae have acquired plasmids that contain AmpC β-lactamase genes.

For AmpC-expressing strains, carbapenems are an appropriate treatment option, especially for severely ill patients. Meta-analyses support piperacillin-tazobactam as a possible option. The fourth-generation cephalosporin cefepime may be an appropriate option if the concomitant presence of an ESBL can be excluded (a task that currently exceeds the capability of most clinical microbiology laboratories) and source control is achieved. Vaborbactam and avibactam are the most potent β-lactamase inhibitors. Ceftazidime-avibactam, imipenem/cilastatin-relebactam, and cefiderocol are active in vitro, but clinical data are limited. Other carbapenem-sparing alternatives to consider if isolates are susceptible in vitro include fluoroquinolones, TMP-SMX, and aminoglycosides. Tigecycline, eravacycline, and omadacycline are active in vitro (except against *Proteus*, *Morganella*, *Providencia*, and *Serratia*).

Carbapenemases of Ambler class A (serine-based hydrolytic mechanism; *K. pneumoniae* carbapenemase [KPC], *Serratia marcescens* enzyme [SME]) and class B (metallo [zinc]-based hydrolytic mechanism; New Delhi metallo-β-lactamase [NDM], Verona integron-mediated metallo-β-lactamase [VIM], active on imipenem [IMP]) confer resistance to the same drugs as do ESBLs, plus to cephamycins and carbapenems. By contrast, Ambler class D carbapenemases (serine-based hydrolytic mechanism, e.g., oxacillinase [OXA]) hydrolyze carbapenems and penicillins, but they have minimal activity against extended-spectrum cephalosporins. As with ESBLs, carbapenemase-encoding genes may be present on transferable plasmids, which often encode linked resistance to fluoroquinolones, TMP-SMX, tetracyclines, and aminoglycosides. Transposon-mediated spread (e.g., TN4401 for KPC) is also important. Although all major carbapenemases have been described around the globe, KPC is most common in the Americas, NDM in Asia, and OXA in Europe. Asymptomatic intestinal carriage may facilitate spread.

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Carbapenemase production by Enterobacteriaceae (CPE) is most prevalent in *K. pneumoniae*, followed by *Enterobacter* spp. and *E. coli*, but has been described in nearly all members of the family. *M. morganii*, *Proteus*, and *Providencia* exhibit intrinsic low-level imipenem resistance. Although for carbapenem-resistant isolates the Clinical and Laboratory Standards Institute (CLSI) no longer recommends routine identification of CPE, such data conceivably could inform epidemiologic surveillance, infection control efforts, antimicrobial stewardship, and treatment decisions, especially if susceptibility data for selected agents are not available. Genotypic and phenotypic methods can detect carbapenemase genes or activity. Each of these methodologies has pros and cons. At the time of this writing, CLSI endorses the modified carbapenem inactivation method and the Carba NP test.

For the treatment of infections due to Enterobacteriaceae that produce class A or D carbapenemases (serine-based hydrolysis; KPC, OXA, SME), ceftazidime-avibactam is emerging as a first-line agent particularly for bacteremia, but suboptimal efficacy has been observed with pneumonia and in patients on renal replacement therapy, and resistance has developed in up to 10% of cases. Polymyxins are also active. Clinical success against KPC-producing CRE has also been reported for meropenem-vaborbactam and, to a lesser extent, imipenem/cilastatin-relebactam; importantly, however, neither of these agents is active against OXA-producing CRE. For SME, based on limited in vitro data, vaborbactam is most active, followed by avibactam, whereas relebactam is significantly less active; this suggests that meropenem-vaborbactam or ceftazidime-avibactam should be viable treatment options for SME-producing CRE. Ceftazidime, cefepime, and aztreonam are active against OXA-48-like–producing CRE.

Treatment of infections due to class B metallo-β-lactamase–producing CRE is more challenging. The polymyxins B and E currently constitute one of the last lines of defense against strains that produce metallo-carbapenemases (e.g., NDM-1). However, these agents' nephrotoxicity and neurotoxicity potential, their limited clinical efficacy, and the recent emergence of the polymyxin resistance gene *mcr-1* on a stable transferable plasmid and *mcr-1*–independent resistance threaten their utility.

Aztreonam is active against metallo-carbapenemases but is hydrolyzed by ESBLs and AmpC β-lactamases, which often coexist in XDR strains. Ongoing studies are assessing aztreonam plus avibactam, a promising combination with in vitro activity against class A, B, and D enzymes, for the treatment of CRE strains that produce both NDM plus KPC or OXA. A currently available workaround involving approved drugs is co-administration of ceftazidime-avibactam and aztreonam; avibactam protects aztreonam from hydrolysis from ESBLs and AmpC β-lactamases.

Although tigecycline, eravacycline, and omadacycline are active in vitro, pharmacokinetic-pharmacodynamic limitations exist, and along with the polymyxins, they exhibit poor activity against the Proteeae and *Serratia*. Cefiderocol is active in vitro against KPC, most NDM carbapenemases, and OXA (i.e., classes A, B, and D enzymes); clinical trials for the treatment of carbapenemase-resistant GNB are in progress. Aminoglycosides, of which plazomicin is most active, may have some utility for combination therapy. Fosfomycin is often active in vitro, but clinical data in the treatment of serious infections due to CPE are limited, resistance may develop with monotherapy, and a parenteral formulation is not yet available in the United States and certain other countries. Collectively, these considerations recommend fosfomycin as a second-line agent and for use in combination therapy, except perhaps for prostatitis due to superior penetration.

Carbapenem resistance in the absence of carbapenemases can occur in the presence of ESBLs or AmpC β-lactamases in combination with porin mutations (non-CP-CRE); however, most laboratories will not be able to differentiate CPE from non-CP-CRE. The non-CP-CRE phenotype is most commonly seen in *E. coli* and *Enterobacter* spp. In general, resistance to noncarbapenem antimicrobial classes is less, but data are limited on the optimal management approach for non-CP-CRE.

β-Lactamase inhibitor resistance is an uncommon (4% of *E. coli*/*K. pneumoniae* blood isolates) but increasingly recognized phenotype that is characterized by resistance to β-lactamase inhibitors, but not to third-generation cephalosporins. This mechanism of resistance is distinct from production of ESBLs, AmpC β-lactamases, and carbapenemases, and is still being delineated. Limited evidence suggests that ceftriaxone is an appropriate treatment option for such strains.

Fluoroquinolone resistance is usually due to alterations in or protection of the target sites in DNA gyrase and topoisomerase IV, with or without decreased permeability and active efflux. Fluoroquinolone resistance is increasingly prevalent among GNB and is associated with resistance to other antimicrobial classes; for example, 20–80% of ESBL-producing enteric GNB are also resistant to fluoroquinolones. At present, fluoroquinolones should be considered unreliable as empirical therapy for GNB infections in critically ill patients.

Aminoglycoside resistance in Enterobacteriaceae is conferred via enzymatic modification by *N*-acetyltransferases, *O*-adenyltransferases, or *O*-phosphotransferases, which in turn affects ribosomal binding. Amikacin is less affected by these transferases than gentamicin and tobramycin and therefore is generally more active. Plazomicin is unaffected by all of these enzymes that confer resistance to amikacin, gentamicin, and tobramycin, thereby making this an important alternative agent in the treatment of selected XDR strains (excepting the Proteeae, against which plazomicin is poorly active). An as yet uncommon resistance mechanism involves 16S ribosomal RNA methylases, which prevent all aminoglycosides (including plazomicin) from binding to their target ribosomes. To date, these methylases are most common in strains that possess metallo-carbapenemases (e.g., NDM).

PREVENTION

(See also Chap. 142) Certain measures are broadly applicable for decreasing infection risk. Antimicrobial stewardship programs should be instituted to facilitate appropriate antimicrobial use, which will minimize the development of resistance. Diligent adherence to hand-hygiene protocols by health care personnel and cleaning/disinfection or single-patient use of objects that come into contact with patients (e.g., stethoscopes and blood pressure cuffs) are essential. Indwelling devices (e.g., urinary and intravascular catheters) should be used only when necessary and inserted according to an appropriate protocol; protocols for daily-use evaluation and prompt removal should be implemented. Multi-use medication vials should be avoided if possible. Oral application of chlorhexidine decreases the incidence of pneumonia among patients on ventilators. Increasing data support the implementation of universal decolonization (e.g., chlorhexidine bathing) to prevent infection in ICU patients. The public health threat from CRE has resulted in additional recommendations, especially for carbapenemase-producing CRE, which are an even greater concern. These recommendations include contact precautions for patients colonized or infected with CRE, notification to the receiving facility from facilities transferring such a patient, and daily environmental cleaning. Screening of contacts and active surveillance for these bacteria may also be appropriate.

ESCHERICHIA COLI INFECTIONS



All *E. coli* strains share a core genome of ~2000 genes. In contrast, an *E. coli* strain's ability to cause infection and the nature of such infections are defined largely by accessory (i.e., noncore, nonessential) genes that encode various virulence factors. The composition of the *E. coli* accessory genome is continuously in flux, as demonstrated by the recent evolution of Shiga toxin–producing enteroaggregative *E. coli*.

COMMENSAL STRAINS

Commensal *E. coli* variants are an important constituent of the normal intestinal microbiota that confer benefits to the host (e.g., resistance to colonization with pathogenic organisms). Such strains generally lack the specialized virulence traits that enable extraintestinal and

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intestinal pathogenic *E. coli* strains to cause disease outside and within the gastrointestinal tract, respectively. However, even commensal *E. coli* strains can be involved in extraintestinal infections in the presence of an aggravating factor, such as a foreign body (e.g., a urinary catheter), host compromise (e.g., local anatomic or functional abnormalities [including urinary or biliary tract obstruction] or systemic immunocompromise), or an inoculum that is large or contains a mixture of bacterial species (e.g., fecal contamination of the peritoneal cavity).

EXTRAINTESTINAL PATHOGENIC STRAINS

ExPEC strains are the most common enteric GNB to cause community-acquired and health care–associated bacterial infections. The emerging propensity of these strains to acquire new mechanisms of antimicrobial resistance (e.g., FQ resistance mutations, ESBLs, carbapenemases) poses novel challenges in managing ExPEC infection. Several ExPEC clonal groups (e.g., sequence types [STs] ST131, ST95, ST69, ST73) are recognized to have undergone global dissemination. The mechanisms underlying the epidemiologic success of such disseminated lineages remain an area of active investigation. In the case of ST131, efficient human-to-human transmission followed by colonization and long-term persistence within the intestinal microbiota appears to be a critical factor. Although acquisition of ESBL-producing *E. coli* from the food chain has been described, this appears to occur relatively uncommonly.

Like commensal *E. coli* (but unlike intestinal pathogenic *E. coli*), ExPEC strains are often found in the intestinal microbiota of healthy individuals and, except for rare chimeric ExPEC/intestinal pathogenic *E. coli* strains, do not cause gastroenteritis in humans. Entry from their site of colonization (e.g., the colon, vagina, or oropharynx) into a normally sterile extraintestinal site (e.g., the urinary tract, peritoneal cavity, or lungs) is the rate-limiting step for infection. ExPEC strains have acquired accessory genes encoding diverse virulence factors that enable the bacteria to cause infections outside the gastrointestinal tract in both normal and compromised hosts (Table 161-1). These virulence genes define ExPEC and, for the most part, are distinct from the virulence genes that enable intestinal pathogenic strains to cause diarrheal disease **(Table 161-3).** All age groups, all types of hosts, and nearly all organs and anatomic sites are susceptible to infection by ExPEC. Even previously healthy hosts can become severely ill or die when infected with ExPEC; however, adverse outcomes are more common among hosts with comorbid illnesses and host defense abnormalities. The diversity and the medical and economic impact of ExPEC infections are evident from consideration of the following specific syndromes.

Extraintestinal Infectious Syndromes

• URINARY TRACT INFECTION The urinary tract is the site most frequently infected by ExPEC. UTI is an exceedingly common infection among ambulatory patients, accounting for 1% of ambulatory care visits in the United States and second only to lower respiratory tract infection among infections responsible for hospitalization. UTIs are best considered by clinical syndrome (e.g., cystitis, pyelonephritis, catheter-associated UTI) and within the context of specific hosts (e.g., premenopausal women, compromised hosts; **Chap. 135**). *E. coli* is the single most common pathogen for all UTI syndrome/host group combinations. Each year in the United States, *E. coli* causes 80–90% of the estimated 6–8 million episodes of cystitis that occur in ambulatory, premenopausal women with an anatomically and functionally normal urinary tract (i.e., uncomplicated cystitis). Furthermore, 20% of women with an initial cystitis episode develop frequent recurrences.

Uncomplicated cystitis, the most common acute UTI syndrome, is characterized by dysuria, urinary frequency and urgency, and suprapubic pain. Progression to more severe infection is rare; the natural history is slow spontaneous symptom resolution, which antimicrobial therapy hastens. Fever and/or back pain suggest progression to pyelonephritis. Even when pyelonephritis is treated effectively, fever may take 5–7 days to resolve completely. Persistently elevated or increasing fever, flank pain, and neutrophil counts should prompt evaluation for intrarenal or perinephric abscess and/or obstruction. Pyelonephritis uncommonly causes renal parenchymal damage and loss of renal function, primarily in association with urinary obstruction, which can be preexisting or, rarely, occurs de novo in diabetic patients who develop renal papillary necrosis as a result of kidney infection. Pregnant women are at unusually high risk for developing pyelonephritis, which can adversely affect the outcome of pregnancy. As a result, prenatal screening for and treatment of asymptomatic bacteriuria during pregnancy are standard. Prostatic infection (prostatitis), a potential complication of UTI in men, can present either acutely (severe), which is rare, or in a chronic manner (recurrent cystitis), which is much more common. Acute pyelonephritis, acute prostatitis, and other systemic illnesses due to UTI can be designated collectively as *urosepsis*, *febrile UTI*, or *systemic UTI*, and may or may not be accompanied by bacteremia. The diagnosis and treatment of UTI, as detailed in **Chap. 135**, should be tailored to the individual host, the nature and site of infection, and local patterns of antimicrobial susceptibility.

ABDOMINAL AND PELVIC INFECTION The abdomen/pelvis is the second most common site of extraintestinal infection due to *E. coli*. A wide variety of clinical syndromes occur in this location, including acute peritonitis secondary to fecal contamination, spontaneous bacterial peritonitis, dialysis-associated peritonitis, diverticulitis, appendicitis, intraperitoneal or visceral abscesses (hepatic, pancreatic, splenic), infected pancreatic pseudocysts, and septic cholangitis and/or cholecystitis. In intraabdominal infections, *E. coli* can be isolated either alone or, as occurs more often, in combination with other facultative and/or anaerobic members of the intestinal microbiota **(Chap. 132)**.

PNEUMONIA *E. coli* is not usually considered an important cause of pneumonia **(Chap. 126)**. Indeed, enteric GNB account for only 1–3%

TABLE 161-3 Intestinal Pathogenic *Escherichia coli*

PATHOTYPE	EPIDEMIOLOGY	CLINICAL SYNDROME[a]	DEFINING MOLECULAR TRAIT	RESPONSIBLE GENETIC ELEMENT[b]
STEC/EHEC/ ST-EAEC	Food, water, person-to-person; all ages, industrialized countries	Hemorrhagic colitis, hemolytic-uremic syndrome	Shiga toxin	Lambda-like Stx1- or Stx2-encoding bacteriophage
ETEC	Food, water; young children in and travelers to developing countries	Traveler's diarrhea	Heat-stable and labile enterotoxins, colonization factors	Virulence plasmid(s)
EPEC	Person-to-person; young children and neonates in developing countries	Watery diarrhea, persistent diarrhea	Localized adherence, attaching and effacing lesion on intestinal epithelium	EPEC adherence factor plasmid pathogenicity island (locus for enterocyte effacement [LEE])
EIEC	Food, water; children in and travelers to developing countries	Watery diarrhea, occasionally dysentery	Invasion of colonic epithelial cells, intracellular multiplication, cell-to-cell spread	Multiple genes contained primarily in a large virulence plasmid
EAEC	?Food, water; children in and travelers to developing countries; all ages, industrialized countries	Traveler's diarrhea, acute diarrhea, persistent diarrhea	Aggregative/diffuse adherence, virulence factors regulated by AggR	Chromosomal or plasmid-associated adherence and toxin genes

[a]Classic syndromes; see text for details on disease spectrum. [b]Pathogenesis involves multiple genes, including genes in addition to those listed.

Abbreviations: EAEC, enteroaggregative *E. coli;* EHEC, enterohemorrhagic *E. coli*; EIEC, enteroinvasive *E. coli*; EPEC, enteropathogenic *E. coli*; ETEC, enterotoxigenic *E. coli*; ST-EAEC, Shiga toxin–producing enteroaggregative *E. coli*; STEC, Shiga toxin–producing *E. coli.*

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of cases of community-acquired pneumonia, in part because these organisms colonize the oropharynx only transiently in a minority of healthy individuals. However, rates of oral colonization with *E. coli* and other GNB increase with severity of illness and antibiotic use. Consequently, GNB are a more common cause of pneumonia among residents of LTCFs and are the most common cause (60–70% of cases) of hospital-acquired pneumonia **(Chap. 142)**, particularly among postoperative and ICU patients (e.g., ventilator-associated pneumonia).

Pulmonary infection is usually acquired by small-volume aspiration but occasionally occurs via hematogenous spread, in which case multifocal nodular infiltrates can be seen. Tissue necrosis, probably due in part to bacterial cytotoxins, is common. Despite significant institutional variation, *E. coli* is generally the third or fourth most commonly isolated type of GNB in hospital-acquired pneumonia, accounting for 5–8% of episodes in both U.S.-based and Europe-based studies. Regardless of the host, pneumonia due to ExPEC is a serious disease, with high crude and attributable mortality rates (20–60% and 10–20%, respectively).

MENINGITIS **(See also Chap. 138)** *E. coli* is one of the leading causes of neonatal meningitis, together with group B *Streptococcus*. Most *E. coli* strains that cause neonatal meningitis possess the K1 capsular antigen and derive from a limited number of meningitis-associated clonal groups. Ventriculomegaly occurs commonly. After the first month of life, *E. coli* meningitis is uncommon, and usually accompanies surgical or traumatic disruption of the meninges or hepatic cirrhosis. In patients with cirrhosis who develop meningitis, the meninges are presumably seeded due to poor hepatic clearance of portal vein bacteremia.

CELLULITIS/MUSCULOSKELETAL INFECTION *E. coli* contributes frequently to infections of decubitus ulcers and occasionally to infections of lower-extremity ulcers and wounds in diabetic patients and other hosts with neurovascular compromise. Osteomyelitis secondary to contiguous spread can occur in these settings. *E. coli* also causes cellulitis or infections of burn sites and surgical wounds (accounting for ~10% of surgical site infections), particularly when the infection originates close to the perineum. *E. coli* causes hematogenously acquired osteomyelitis, especially of vertebral discs and bodies, accounting for up to 10% of cases in some series **(Chap. 131)**. *E. coli* occasionally causes orthopedic device–associated infection or septic arthritis and rarely causes hematogenous myositis. Myositis or fasciitis of the thigh due to *E. coli* should prompt an evaluation for an abdominal source with contiguous spread.

ENDOVASCULAR INFECTION Despite being one of the most common causes of bacteremia, *E. coli* rarely seeds native heart valves. When the organism does infect native valves, it usually does so in the setting of prior valvular disease. *E. coli* infections of aneurysms, the portal vein (*pylephlebitis*), and vascular grafts are quite uncommon.

MISCELLANEOUS INFECTIONS *E. coli* can cause infection in nearly every organ and anatomic site. It occasionally causes postoperative mediastinitis or complicated sinusitis and uncommonly causes endophthalmitis, ecthyma gangrenosum, or brain abscess.

BACTEREMIA *E. coli* bacteremia can arise from infection at any extraintestinal site. In addition, *E. coli* bacteremia can arise from percutaneous intravascular devices, transrectal prostate biopsy, and the increased intestinal mucosal permeability seen in neonates and patients with advanced cirrhosis, neutropenia, chemotherapy-induced mucositis, trauma, and extensive burns. *E. coli* bacteremia due to an ESBL-producing strain also has been reported after fecal microbiota transplant in patients with increased mucosal permeability. Roughly equal proportions of *E. coli* bacteremia cases originate in the community and in health care settings. Isolation of *E. coli* from the blood is almost always clinically significant and may be accompanied by the sepsis syndrome (dysfunction of at least one organ or system) or septic shock **(Chap. 304)**.

The urinary tract is the most common source for *E. coli* bacteremia, accounting for one-half to two-thirds of episodes. Bacteremia from a urinary tract source is particularly common among patients with pyelonephritis, urinary tract obstruction, or urinary instrumentation in the presence of infected urine. The abdomen is the second most common source, accounting for ~25% of episodes. Although many of these episodes result from biliary obstruction (stones, tumor) and overt bowel disruption, which typically are readily apparent, some abdominal sources (e.g., abscesses) are remarkably silent clinically and require identification via imaging studies (e.g., computed tomography). Therefore, especially given the high prevalence of asymptomatic bacteriuria among elderly and functionally compromised individuals, the physician should be cautious in attributing *E. coli* bacteremia to a urinary source in the absence of characteristic signs and symptoms of UTI. Soft tissue, bone, pulmonary infections, and intravascular catheter infections are other sources of *E. coli* bacteremia.

Diagnosis Strains of *E. coli* that cause extraintestinal infections usually grow both aerobically and anaerobically within 24 h on standard diagnostic media and are identified readily by the clinical microbiology laboratory according to routine biochemical criteria. More than 90% of ExPEC strains are rapid lactose fermenters and are indole-positive.

TREATMENT

Extraintestinal *E. coli* Infections

E. coli does not possess clinically significant intrinsic resistance to antimicrobials; however, increasing acquired resistance is making treatment problematic. Although geographic differences exist, in general, the prevalence of resistance is >20% for ampicillin, amoxicillin-clavulanate, ampicillin-sulbactam, cefazolin, TMP-SMX, and fluoroquinolones, even in community-acquired infections. This resistance precludes empirical use of these agents for serious infections. Travel outside of the United States, prior exposure to an antimicrobial agent, or exposure to a health care setting further increases the likelihood of resistance. Fortunately, >90% of isolates that cause uncomplicated cystitis remain susceptible to nitrofurantoin and fosfomycin.

From 2015 to 2017, the U.S. National Healthcare Safety Network (USNHSN) identified 24% of *E. coli* clinical isolates as ESBL-producers. Higher prevalences are reported from Asia, Eastern Europe, South America, and Africa; prevalence is also greater in isolates from health care settings, especially LTCFs. However, community-acquired UTIs caused by *E. coli* strains that produce CTX-M ESBLs are increasingly common. Oral treatment options for such strains are limited; however, in vitro and limited clinical data indicate that fosfomycin, pivmecillinam, and nitrofurantoin are highly active and can be used for cystitis, and omadacycline is predicted to be active based on in vitro data. For parenteral therapy of carbapenem-sensitive strains, the most predictably active agents (>90%) include carbapenems, amikacin, plazomicin, ceftazidime-avibactam, ceftolozane-tazobactam, meropenem-vaborbactam, imipenem/cilastatin-relebactam, piperacillin-tazobactam, polymyxins, cefiderocol, tigecycline, eravacycline, and omadacycline. Treatment of carbapenemase-producing strains is dependent of the class of enzyme produced (see "Carbapenemase" above). Uncertainty exists on the optimal treatment for non-CP-CR *E. coli*.

Empirical treatment decisions for critically ill patients should be dictated by local susceptibility patterns and patient-specific risk factors (1.2% prevalence from the USNHSN 2015–2017 data). Equally important as prompt institution of effective empirical therapy for seriously ill patients is use of appropriate narrower-spectrum agents for definitive therapy whenever possible and avoidance of treatment for patients who are colonized but not infected.

INTESTINAL PATHOGENIC STRAINS

Pathotypes Certain strains of *E. coli* are capable of causing diarrheal disease. **(Other important intestinal pathogens are discussed in Chaps. 133, 134, and 165–168.)** At least in the industrialized world, intestinal pathogenic *E. coli* strains are rarely encountered in the fecal flora of healthy persons, and instead appear to be essentially obligate

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pathogens. These strains have evolved a special ability to cause enteritis, enterocolitis, and colitis when ingested in sufficient quantities by a naive host. At least five distinct pathotypes of intestinal pathogenic *E. coli* exist: (1) Shiga toxin–producing *E. coli* (STEC), which includes the subsets enterohemorrhagic *E. coli* (EHEC) and the recently evolved Shiga toxin–producing enteroaggregative *E. coli* (ST-EAEC); (2) enterotoxigenic *E. coli* (ETEC); (3) enteropathogenic *E. coli* (EPEC); (4) enteroinvasive *E. coli* (EIEC); and (5) enteroaggregative *E. coli* (EAEC). Diffusely adherent *E. coli* (DAEC) and cytodetaching *E. coli* are additional putative pathotypes. Lastly, a variant termed adherent invasive *E. coli* (AIEC) has been associated with Crohn's disease (although a causal role remains unproven) but does not cause acute diarrheal disease.

Contaminated food and water are the primary transmission vehicles for ETEC, STEC/EHEC/ST-EAEC, EIEC, and EAEC, whereas person-to-person spread (direct or indirect) is the primary transmission route for EPEC and a secondary transmission route for STEC/EHEC/ST-EAEC. Gastric acidity confers some protection against infection; therefore, persons with decreased stomach acid levels are especially susceptible. Humans are the major reservoir for such strains (except for STEC/EHEC, for which bovines are the main carriers); host range appears to be dictated by species-specific attachment factors. Although some overlap exists, each pathotype possesses a distinctive combination of virulence traits that results in a pathotype-specific pathogenic mechanism (Table 161-3). With rare exceptions, these strains are largely incapable of causing disease outside the intestinal tract. Whereas disease due to STEC/EHEC/ST-EAEC occurs primarily in high-income countries, disease due to ETEC, EPEC, and EIEC occurs primarily in low- and middle-income countries in Asia, Africa, and Latin America, and disease due to EAEC occurs globally.

SHIGA TOXIN–PRODUCING *E. COLI* STEC/EHEC/ST-EAEC strains are pathogens that can cause hemorrhagic colitis and the hemolytic-uremic syndrome (HUS). In contrast to other intestinal pathotypes, STEC/EHEC/ST-EAEC causes infections more frequently in industrialized countries than in developing regions. Several large outbreaks resulting from the consumption of fresh produce (e.g., lettuce, spinach, sprouts) and of undercooked ground beef have received significant media attention. In addition, a dramatic 2011 outbreak—mainly in Germany—involved an EAEC strain that acquired a Shiga toxin–encoding phage, resulting in a novel genotype, ST-EAEC (O104:H4). This strain was transmitted to the primary cases by sprouted fenugreek seeds, with subsequent human-to-human transmission, and resulted in >4000 cases and 54 deaths.

STEC strains are the fourth most commonly reported cause of bacterial diarrhea in the United States (after *Campylobacter*, *Salmonella*, and *Shigella*). O157:H7 is the most prominent serotype among STEC strains, but many other serogroups have been described, including O6, O26, O45, O55, O91, O103, O111, O113, O121, and O145. Domesticated ruminant animals, particularly cattle and young calves, serve as the major reservoir for STEC/EHEC. Ground or mechanically tenderized beef—the most common food source of STEC/EHEC strains—is often contaminated with intestinal bacteria from the source animals during processing. Furthermore, manure from cattle or other animals (including in the form of fertilizer) can contaminate produce (potatoes, lettuce, spinach, sprouts, fallen fruits, nuts, strawberries), and fecal runoff from manure can contaminate water systems. Dairy products and petting zoos are additional sources of infection.

It is estimated that $<10^2$ colony-forming units (CFU) of STEC/EHEC/ST-EAEC can cause disease. Therefore, not only can low levels of food or environmental contamination (e.g., in water swallowed while swimming) result in disease, but person-to-person transmission (e.g., at day-care centers and in institutions) is an important route for secondary spread. Laboratory-associated infections also occur. Illness due to this group of pathogens peaks in the summer months and occurs both as outbreaks and as sporadic cases.



For STEC/EHEC/ST-EAEC, production of Shiga toxin (Stx2a-g and/or Stx1a,c,d) is a critical factor for occurrence of clinical disease, as demonstrated by the 2011 ST-EAEC outbreak. The *stx* gene is present on chromosomally integrated prophages, and various combinations of *stx* types and subtypes can occur in a given strain. *Shigella dysenteriae* strains that produce the closely related Shiga toxin Stx also can cause hemorrhagic colitis and HUS. Stx2 (especially Stx2a,c,d) appears to be more important than Stx1 in the development of HUS. All Shiga toxins studied to date are multimers; they comprise one A subunit that is enzymatically active and five identical B subunits that mediate binding to globosyl ceramides, which are membrane-associated glycolipids expressed on certain host cells. As in ricin, the Stx A subunit cleaves an adenine from the host cell's 28S rRNA, thereby irreversibly inhibiting ribosomal function (i.e., protein synthesis) and potentially leading to apoptosis.

For full pathogenicity, STEC strains require additional properties such as acid tolerance and epithelial cell adherence. Most disease-causing isolates possess the chromosomal locus for enterocyte effacement (LEE). This pathogenicity island was first described in EPEC strains; it contains genes that mediate adherence to intestinal epithelial cells and a system that subverts host cells by the translocation of bacterial proteins (type III secretion system). EHEC strains make up the subgroup of STEC strains that possess stx_1 and/or stx_2, as well as LEE. In contrast, the 2011 ST-EAEC outbreak strain lacked LEE, yet was associated with a higher proportion of patients developing HUS (22%) than the historical average for STEC/EHEC outbreaks (2–8%). Data support the essential role of the 2011 outbreak strain's EAEC-associated virulence factors (e.g., AAF/I fimbriae, serine proteases SigA, SepA) in adherence, increased inflammation, and disruption of the intestinal epithelial barrier, which in turn increased the systemic translocation of Stx2a.

After exposure to STEC/EHEC/ST-EAEC and a 3- to 4-day incubation period, colonization of the colon and perhaps the ileum results in symptoms. Colonic edema and an initial nonbloody secretory diarrhea may progress to the hallmark syndrome of grossly bloody diarrhea (identified by history or examination). Significant abdominal pain and fecal leukocytes are common (70% of cases), whereas fever is not; absence of fever can incorrectly lead to consideration of noninfectious conditions (e.g., intussusception and inflammatory or ischemic bowel disease). Occasionally, infections caused by *C. difficile*, *K. oxytoca* (see "*Klebsiella* Infections," below), *Campylobacter*, and *Salmonella* present in a similar fashion. STEC/EHEC disease is usually self-limited, lasting 5–10 days.

A feared complication of infection with STEC/EHEC strains is HUS, which occurs 2–14 days after diarrhea, most often in young children (estimated to occur in 15% of infected children <10 years of age) or elderly patients. It is estimated that in the United States >50% of all HUS cases—and 90% of HUS cases in children, which is a leading cause of acute renal failure in this latter population—are caused by STEC/EHEC. In contrast, with ST-EAEC infection, HUS occurs more commonly among nonelderly adults, especially young women. HUS is mediated by the systemic translocation of Shiga toxins. Erythrocytes may serve as carriers of Stx to endothelial cells located in the small vessels of the kidney and brain. The subsequent development of thrombotic microangiopathy (perhaps with direct toxin-mediated effects on various nonendothelial cells) commonly produces some combination of fever, hemolytic anemia, thrombocytopenia, renal failure, and encephalopathy. Stx-mediated complement activation may also play a role in the development of HUS. Although with dialysis support the mortality rate of HUS is <10%, survivors often have persisting renal and neurologic dysfunction.

ENTEROTOXIGENIC *E. COLI* ETEC is a major cause of endemic diarrhea in low- and middle-income countries and is responsible for an estimated 800 million cases annually. After weaning, children in these locales commonly experience several episodes of ETEC infection during the first 3 years of life. The incidence of disease diminishes with age, a pattern that correlates with the development of mucosal immunity to colonization factors (i.e., adhesins).

In industrialized countries, ETEC is the most common agent of traveler's diarrhea, causing 25–75% of cases. The incidence of infection may be decreased by prudent avoidance of potentially contaminated fluids and foods, particularly items that are raw, insufficiently cooked, peeled, or unrefrigerated **(Chap. 124)**. ETEC infection is uncommon

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in the United States, but outbreaks secondary to consumption of food products imported from endemic areas have occurred. A large inoculum (10^6–10^8 CFU) is needed to produce disease, which usually develops after an incubation period of 12–72 h.



After adherence of ETEC to enterocytes via colonization factors (e.g., CFA/I, CS), disease is mediated, primarily by a heat-labile toxin (LT) and/or a heat-stable toxin (STa), leading to diarrheal disease. Disease is less severe with strains that produce only LT. Both LT and STa cause net fluid secretion via activation of adenylate cyclase and/or guanylate cyclase C (STa) in the jejunum and ileum. The result is watery diarrhea accompanied by cramps.

LT consists of an A and a pentameric B subunit and is structurally and functionally similar to cholera toxin. Strong binding of the B subunit to the GM_1 ganglioside on intestinal epithelial cells leads to the intracellular translocation of the A subunit, which functions as an ADP-ribosyltransferase. Mature STa is an 18- or 19-amino-acid secreted peptide that leads to increased intracellular concentrations of cGMP. Characteristically absent in ETEC-mediated disease are histopathologic changes within the small bowel; mucus, blood, and inflammatory cells in stool; and fever.

The disease spectrum of ETEC infection ranges from mild illness to a life-threatening, cholera-like syndrome. Although symptoms are usually self-limited (typically lasting for 3–5 days), infection may result in significant morbidity and mortality (>250,000 deaths annually, mostly from profound volume depletion) when access to health care or suitable rehydration fluids is limited and when small and/or undernourished children are affected.

ENTEROPATHOGENIC *E. COLI* EPEC causes disease primarily in young children, including neonates. The first *E. coli* pathotype recognized as an agent of diarrheal disease, EPEC was responsible for outbreaks of infantile diarrhea (including in hospital nurseries) in industrialized countries in the 1940s and 1950s. At present, EPEC infection is uncommon in high-income countries, but among infants in low- and middle-income countries, it is an important cause of diarrhea (both sporadic and epidemic), often accompanied by vomiting and fever. Breast-feeding diminishes the incidence of EPEC infection. Rapid person-to-person spread may occur.



Symptoms develop after colonization of the small bowel and a brief incubation period (1 or 2 days). Initial localized adherence to enterocytes via type IV bundle-forming pili leads to a characteristic effacement of microvilli, with the formation of cuplike, actin-rich pedestals mediated by factors in the LEE. Diarrhea production is a complex and regulated process in which host cell modulation by a type III secretion system plays an important role. Strains lacking bundle-forming pili have been categorized as atypical EPEC (aEPEC); increasing data support a role for these strains as intestinal pathogens in all age groups and among HIV-infected individuals. Diarrheal stool often contain mucus but not blood. Although EPEC diarrhea is usually self-limited (lasting 5–15 days), it may persist for weeks.

ENTEROINVASIVE *E. COLI* EIEC, a relatively uncommon (or perhaps underrecognized) cause of diarrhea, is rarely identified in the United States, although a few food-related outbreaks have been described. In low- and middle-income countries, sporadic disease is recognized infrequently in children and travelers.

EIEC shares many genetic and clinical features, as well as a common ancestor, with *Shigella*. Both are intracellular pathogens for which virulence is mediated by the presence of specific factors and by the loss or inactivation of other factors (antivirulence genes), which presumably occurred during these organisms' transition from an extracellular to an intracellular lifestyle.

Colonization and invasion of the colonic mucosa, followed by replication therein and cell-to-cell spread (in part via a type III secretion system), result in the development of inflammatory colitis. However, unlike *Shigella*, EIEC produces disease only with a large inoculum (10^8–10^{10} CFU) and is less virulent, typically causing only mild, self-limited (7–10 days), watery diarrhea. Onset generally follows an incubation period of 1–3 days. Occasionally, EIEC can cause a shigellosis-like (dysentery) syndrome characterized by fever, abdominal pain, tenesmus, and scant stool containing mucus, blood, and inflammatory cells.

ENTEROAGGREGATIVE AND DIFFUSELY ADHERENT *E. COLI* EAEC has been described primarily in low- and middle-income countries and in young children. However, recent studies indicate that it also may be a relatively common cause of diarrhea in all age groups in industrialized countries. EAEC has been recognized increasingly as an important cause of traveler's diarrhea. It is highly adapted to humans—the probable reservoir. A large inoculum is required for infection, which usually manifests as watery and sometimes persistent diarrhea in healthy but also malnourished or HIV-infected hosts.



In vitro, EAEC cells exhibit a diffuse or "stacked-brick" pattern of adherence to small-intestine epithelial cells. Virulence factors that probably are necessary for disease are regulated in large part by the transcriptional activator AggR. The pathogenesis of EAEC disease begins with intestinal adherence, which results from stimulation of epithelial mucus production and bacterial biofilm formation, the latter mediated by fimbriae and possibly the mucinase Pic and dispersin. Inflammation ensues, resulting in epithelial cell exfoliation and intestinal secretion, which is mediated by the enterotoxins Pet, EAST-1, ShET1, and HlyE.

An additional enteric pathotype, DAEC, is associated with diarrheal disease, primarily in children 2–6 years of age in some developing countries, and may cause traveler's diarrhea. The Afa/Dr adhesins may contribute to the pathogenesis of such infections.

Diagnosis Acute infectious diarrhea can be classified as noninflammatory (most commonly viral) or inflammatory (usually bacterial); the latter is suggested by grossly bloody or mucoid stools or a positive test for fecal leukocytes, lactoferrin, or calprotectin **(Chap. 133)**. ETEC, EPEC, DAEC, and EAEC cause noninflammatory diarrhea. Identification of these agents can be achieved with a commercial platform (e.g., BioFire® Film Array® Gastrointestinal Panel can detect STEC, ETEC, EPEC, EAEC, and EIEC). However, organism identification is rarely needed because the associated diseases are self-limited. ETEC causes the majority and EAEC a minority of cases of noninflammatory traveler's diarrhea; here again, however, definitive diagnosis generally is not necessary for management (as discussed below). If diarrhea persists for >10 days despite treatment, *Giardia* or *Cryptosporidium* (or, in immunocompromised hosts, certain opportunistic pathogens) should be sought.

Because of the considerable public-health importance of STEC/EHEC/ST-EAEC infections, including the threat of HUS, the CDC now recommends that all patients with community-acquired diarrhea, whether inflammatory or not, be evaluated for these pathogens by simultaneous culture (to provide an isolate for strain typing and for outbreak detection and control) and detection of Shiga toxin or the corresponding genes. The rationale for testing all cases of community-acquired diarrhea, regardless of clinical features, is that bloody stool and fecal white blood cells (or lactoferrin) are not reliably present with STEC/EHEC/ST-EAEC infection. In addition, the use of both tests increases diagnostic sensitivity over that with either test alone.

O157 STEC/EHEC may be identified via culture by screening for *E. coli* strains that do not ferment sorbitol, with subsequent serotyping and testing for Shiga toxin. Selective or screening media are not available for culture-based detection of non-O157 STEC/EHEC/ST-EAEC strains. Detection of Shiga toxins or toxin genes via DNA-based, enzyme-linked immunosorbent, and cytotoxicity assays offers the advantages of rapidity and detection of non-O157 STEC/EHEC/ST-EAEC strains. Specimens positive for toxin but culture-negative for O157 should be forwarded to the local or state public-health laboratory for specialized testing.

TREATMENT

Intestinal *E. coli* Infections

The mainstay of treatment for all diarrheal syndromes is replacement of water and electrolytes. This measure is especially important for STEC/EHEC/ST-EAEC infection because appropriate volume expansion may protect against renal injury and improve outcome.

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The use of prophylactic antibiotics to prevent traveler's diarrhea generally should be discouraged, especially in light of high rates of antimicrobial resistance. However, in selected patients (e.g., those who cannot afford a brief illness or are predisposed to infection), the use of rifaximin, which is nonabsorbable and is well tolerated, is reasonable.

When stools are free of mucus and blood, early patient-initiated treatment of traveler's diarrhea with a fluoroquinolone or azithromycin decreases the duration of illness, and the use of loperamide may halt symptoms within a few hours. Although dysentery caused by EIEC is self-limited, antimicrobial therapy hastens the resolution of symptoms, particularly in severe cases. In contrast, antimicrobial therapy for STEC/EHEC/ST-EAEC infection (the presence of which is suggested by grossly bloody diarrhea without fever) should be avoided because antibiotics may increase the incidence of HUS (possibly via increased production/release of Stx). In the treatment of HUS, plasmapheresis has no benefit and the value of inhibition of C5 (via eculizumab) is unresolved.

KLEBSIELLA INFECTIONS

K. pneumoniae is the most important *Klebsiella* species from a medical standpoint, causing community-acquired, LTCF-acquired, and nosocomial infections. *K. oxytoca* and *K.* (formerly *Enterobacter*) *aerogenes* are primarily pathogens in LTCFs and hospitals. *Klebsiella* species are broadly prevalent in the environment and colonize the mucosal surfaces of mammals. In healthy humans, the prevalence of *K. pneumoniae* colonization is 5–35% in the colon and 1–5% in the oropharynx; skin is usually colonized only transiently.

Most *Klebsiella* infections in Western countries are caused by "classical" *K. pneumoniae* (cKp) and occur in hospitals and LTCFs. The most common clinical syndromes due to cKp are pneumonia, UTI, abdominal infection, intravascular device infection, surgical site infection, soft tissue infection, and secondary bacteremia. cKp strains have gained notoriety because of their propensity for acquiring treatment-confounding antimicrobial resistance determinants and causing both localized and widespread outbreaks, such as with the global spread of NDM-1-producing cKp strains from India associated with medical tourism. Clonal groups STs 11, 15, 101, 307, and 258/512, many members of which produce carbapenemases, are undergoing international dissemination. Transmission within or between institutions is common. *K. pneumoniae* is nearly four-fold more transmissible than *E. coli*, and, disconcertingly, carbapenemase-producing strains are associated with increased spread compared to carbapenem-susceptible strains.

In addition, hypervirulent *K. pneumoniae* (hvKp) strains that are phenotypically and clinically distinct from cKp have emerged recently, after their initial recognition in Taiwan in 1986. Although hvKp infections have occurred globally in all ethnic groups, most cases have been reported in individuals of Asian ethnicity residing in countries from the Asian Pacific Rim, but also in Asians living in other countries. Affected individuals often have diabetes mellitus. These demographics raise the possibility of a locale-specific distribution of the organism or an increased susceptibility of Asian hosts, especially those who are diabetic. In contrast to the usual health care–associated context for cKp infections in the West, hvKp is capable of causing serious life- and organ-threatening infections in younger, healthy individuals from the community and can spread metastatically from the primary site of infection or present with multiple sites of infection. Of concern, recent reports from Asian countries have demonstrated that hvKp is responsible for an increasing number of health care–associated or hospital-acquired infections.

hvKp infection initially was characterized and distinguished from traditional infections caused by cKp strains by its (1) presentation as community-acquired monomicrobial pyogenic liver abscess (**Fig. 161-1, *top***), (2) occurrence in patients lacking a history of hepatobiliary disease, and (3) propensity for metastatic spread to distant sites. Subsequently, the hvKp pathotype has been recognized as the cause of extrahepatic abscesses and infections with or without liver involvement, including pneumonia; meningitis (in the absence of trauma or

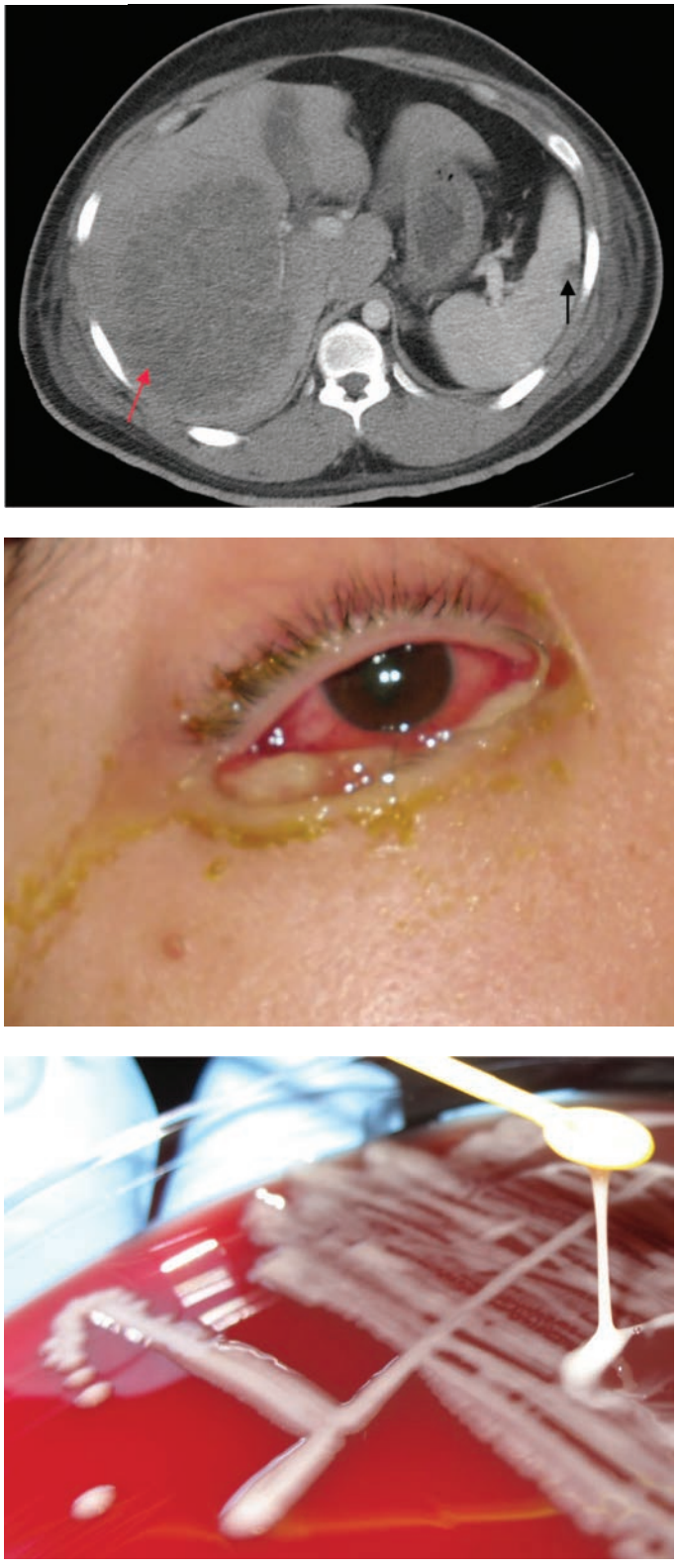


FIGURE 161-1 Hypervirulent pathotype of *K. pneumoniae* (hvKp). ***Top:*** Abdominal CT scan of a previously healthy 24-year-old Vietnamese man shows a primary liver abscess (*red arrow*) with metastatic spread to the spleen (*black arrow*). *(Courtesy of Drs. Chiu-Bin Hsaio and Diana Pomakova.)* ***Middle:*** A previously healthy 33-year-old Chinese man presented with endophthalmitis. *(AS Shon, RP Bajwa, TA Russo: Hypervirulent (hypermucoviscous) Klebsiella pneumoniae: A new and dangerous breed. Virulence 4:107, 2013.)* ***Bottom:*** A hypermucoviscous phenotype (which does not necessarily equate with a mucoid phenotype) has been associated with hvKp strains. A positive string test is shown. However, this test is not optimally sensitive or specific. Identification of the combination of the biomarkers *iucA*, *iroB*, *peg-344*, *rmpA*, and *rmpA2* is presently the most accurate means to identify hvKp.

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neurosurgery); endophthalmitis (Fig. 161-1, *middle*); splenic, psoas, prostatic, epidural, and brain abscesses; and necrotizing fasciitis. Survivors often suffer catastrophic morbidity, such as vision loss and major neurologic sequelae. Most recently, clinicians are faced with an even greater challenge—the confluence of antimicrobial resistance determinants possessed by cKp and the virulence factors possessed by hvKp on the same or coexisting plasmids. The result is the evolution of MDR and XDR hvKp.

K. pneumoniae subspecies *rhinoscleromatis* is the causative agent of rhinoscleroma, a granulomatous mucosal upper-respiratory infection that progresses slowly (over months or years) and causes necrosis and occasionally obstruction of the nasal passages. *K. pneumoniae* subspecies *ozaenae* has been implicated as a cause of chronic atrophic rhinitis and rarely of invasive disease in compromised hosts. *K.* (*Calymmatobacterium*) *granulomatis*, a sexually transmitted pathogen, is the causative agent of granuloma inguinale (donovanosis) that results in chronic genital ulcers (Chap. 173). These *Klebsiella* pathotypes are usually isolated from patients in tropical climates and are genomically distinct from both cKp and hvKp.

INFECTIOUS SYNDROMES

Pneumonia Although cKp accounts for only a small proportion of cases of community-acquired pneumonia in Western countries (Chap. 126), cKP and *K. oxytoca* are common causes of pneumonia among LTCF residents and hospitalized patients because of increased rates of oropharyngeal colonization with these organisms in such individuals. Mechanical ventilation is an important risk factor. In Asia and South Africa, community-acquired pneumonia due to hvKp is becoming increasingly common, rivaling *Streptococcus pneumoniae*, and may occur in younger patients with no underlying disease. *Klebsiella* is also a common cause of pneumonia in severely malnourished children in developing countries.

As in all pneumonias due to enteric GNB, typical manifestations include production of purulent sputum and evidence of airspace disease. Presentation with earlier, less extensive infection is now more common than is the classically described lobar infiltrate, bulging fissure, and currant-jelly sputum. Pulmonary infection due to hvKp that has spread metastatically (e.g., from a hepatic abscess) usually includes nodular bilateral densities, more commonly in the lower lobes. Pulmonary necrosis, pleural effusion, and empyema can occur with disease progression.

UTI cKP accounts for only 1–2% of UTI episodes among otherwise healthy adults but for 5–17% of episodes of UTI in patients with anatomic and functional abnormalities of the urinary tract, including indwelling urinary catheter use (complicated UTI). UTI due to hvKp presents more commonly as renal or prostatic abscess due to bacteremic spread than as ascending infection from the urethra and bladder.

Abdominal Infection cKp causes a spectrum of abdominal infections similar to that caused by *E. coli* but is less frequently isolated from such infections than is *E. coli*. hvKp is a common cause of monomicrobial community-acquired pyogenic liver abscess; in the Asian Pacific Rim, it has been recovered with steadily increasing frequency over the past two decades, replacing *E. coli* as the most common pathogen causing this syndrome. hvKp also is increasingly described as a cause of spontaneous bacterial peritonitis and splenic abscess.

Other Infections When cKp and *K. oxytoca* cause cellulitis or soft tissue infection, the process most frequently involves devitalized tissue (e.g., decubitus and diabetic ulcers, burn wounds) and immunocompromised hosts. cKp and *K. oxytoca* cause some cases of surgical site infection and nosocomial sinusitis as well as occasional cases of osteomyelitis contiguous to soft tissue infection, nontropical myositis, and meningitis (during the neonatal period and after neurosurgery). By contrast, hvKp has become an important cause of community-acquired monomicrobial necrotizing fasciitis, meningitis, endophthalmitis (Fig. 161-1, *middle*), and abscesses within the brain, subdural space, and epidural space, particularly in the Asian Pacific Rim but also globally. Cytotoxin-producing strains of *K. oxytoca* have been implicated as a cause of non–*C. difficile* antibiotic-associated hemorrhagic colitis.

Bacteremia *Klebsiella* infection at any site can produce bacteremia. Infections of the urinary tract, respiratory tract, and abdomen (especially hepatic abscess) each account for 15–30% of episodes of *Klebsiella* bacteremia. Intravascular device–related infections account for another 5–15% of episodes, and surgical site and miscellaneous infections account for the rest. *Klebsiella* is an occasional cause of sepsis in neonates and of bacteremia in neutropenic patients. However, like enteric GNB in general, *Klebsiella* rarely causes endocarditis or other endovascular infections, although the endocarditis can involve extensive valvular destruction.

DIAGNOSIS

Klebsiellae are readily isolated and identified in the laboratory. These organisms usually ferment lactose, although the subspecies *rhinoscleromatis* and *ozaenae* are nonfermenters and are indole-negative. hvKp usually possesses a hypermucoviscous phenotype (Fig. 161-1, *bottom*), although the sensitivity and specificity of the string test are less than optimal. Identification of the combination of the biomarkers *iucA*, *iroB*, *peg-344*, *rmpA*, and *rmpA2* is presently the most accurate means to identify hvKp, although currently, this test is not routinely available.

TREATMENT

Klebsiella Infections

K. (formerly *Enterobacter*) *aerogenes* has a similar resistance profile to *E. cloacae*, the treatment of which is discussed below. *K. pneumoniae* and *K. oxytoca* have similar antibiotic resistance profiles; both are intrinsically resistant to ampicillin. The prevalence of acquired resistance in *K. pneumoniae* and *K. oxytoca* is generally >30% for amoxicillin-clavulanate, ampicillin-sulbactam, nitrofurantoin, and TMP-SMX and ~10–20% for fluoroquinolones, piperacillin-tazobactam, fosfomycin, and omadacycline.

USNHSN data from 2015–2017 identified 25% of *K. pneumoniae* as ESBL-producing strains; higher rates are reported from Asia, South America, and Africa. Although prevalent ESBL-producing strains are greatest in LTCF, isolates of cKp that produce CTX-M ESBLs are increasingly described from the community. Oral treatment for infection due to ESBL-producing strains is more challenging with *Klebsiella* than with *E. coli* because of the comparatively poor activity of nitrofurantoin, the lesser activity of fosfomycin (~80%), and limited available data regarding pivmecillinam (>80%) and omadacycline (75–100% susceptible for ESBL-producing isolates, but 60% if resistant to tetracycline).

Predictably, the ESBL-driven use of carbapenems has selected for strains of cKp and *K. oxytoca* that express carbapenemases (8–18% based on the study and locale, 8.6% prevalence from 2015–2017 USNHSN data). Treatment can be problematic for such organisms, especially those with a metallo-β-lactamase (e.g., NDM), for which the highest prevalences are in cKp and *K. oxytoca* isolates from Eastern Europe and Asia and among health care–associated isolates. Likewise, hvKp strains from Asia are also increasingly reported to produce ESBLs and carbapenemases.

Treatment options for carbapenem-resistant *Klebsiella* are similar to those described for *E. coli* and depend on the class of carbapenemase produced (see "Carbapenemase" above); consultation with relevant experts is advised. For carbapenem-sensitive strains, the most predictably active agents include carbapenems, amikacin, plazomicin, ceftazidime-avibactam, ceftolozane-tazobactam, meropenem-vaborbactam, imipenem/cilastatin-relebactam, polymyxins, cefiderocol, tigecycline, eravacycline, and omadacycline. Empirical treatment decisions for the critically ill patient should be dictated by local susceptibility patterns and patient-specific risk factors.

PROTEUS INFECTIONS

Proteus species are part of the colonic flora of a wide variety of mammals, birds, fish, and reptiles. The ability of these GNB to generate histamine from contaminated fish has implicated them in the pathogenesis of scombroid (fish) poisoning (Chap. 460).

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P. mirabilis causes 90% of *Proteus* infections, which occur in the community, LTCFs, and hospitals. By contrast, *Proteus vulgaris* and *Proteus penneri* are associated primarily with infections acquired in LTCFs or hospitals. Correspondingly, *P. mirabilis* colonizes healthy humans (prevalence, 50%), whereas *P. vulgaris* and *P. penneri* are isolated primarily from individuals with underlying disease. By far the most common site of *Proteus* infection is the urinary tract, where the principal known urovirulence factors of *Proteus* include adhesins, flagella, IgA-IgG protease, iron acquisition systems, and urease. *Proteus* less commonly causes infection at a variety of other extraintestinal sites.

INFECTIOUS SYNDROMES

UTI *P. mirabilis* causes only 1–2% of UTIs in healthy women, and *Proteus* species collectively cause only 5% of hospital-acquired UTIs. However, *Proteus* is responsible for 10–15% of cases of complicated UTI, primarily those associated with catheterization; indeed, *Proteus* accounts for 20–45% of urine isolates from chronically catheterized patients. This high prevalence is due in part to bacterial production of urease, which hydrolyzes urea to ammonia and results in alkalization of the urine. In alkaline urine, organic and inorganic compounds precipitate, contributing to the formation of struvite and carbonate–apatite crystals, biofilms on catheters, and/or frank calculi. *Proteus* becomes associated with the stones and biofilms; thereafter, it usually cannot be eradicated without removal of the stones or catheter. Over time, staghorn calculi may form within the renal pelvis and lead to obstruction and renal failure. Although biologically plausible, clinical support is lacking for the concept that urine samples exhibiting unexplained alkalinity should be cultured, and that isolation of a *Proteus* species (or other urea-splitting organism) should prompt consideration of an evaluation for urolithiasis.

Other Infections *Proteus* occasionally causes pneumonia (primarily in LTCF residents or hospitalized patients), nosocomial sinusitis, intraabdominal abscesses, biliary tract infection, surgical site infection, soft tissue infection (especially decubitus and diabetic ulcers), and osteomyelitis (primarily contiguous); in rare cases, it causes nontropical myositis. In addition, *Proteus* uncommonly causes neonatal meningitis, with the umbilicus frequently implicated as the source; this disease is often complicated by development of a cerebral abscess. Otogenic brain abscess also occurs.

Bacteremia Most episodes of *Proteus* bacteremia originate from the urinary tract, although intravascular devices and any of the less common sites of *Proteus* infection are also potential sources. Endovascular infection is rare. *Proteus* species are occasional agents of sepsis in neonates and of bacteremia in neutropenic patients.

DIAGNOSIS

Proteus is readily isolated and identified in the laboratory. Most strains are lactose-negative, produce H_2S, and demonstrate characteristic swarming motility on agar plates. *P. mirabilis* and *P. penneri* are indole-negative, whereas *P. vulgaris* is indole-positive. The inability to produce ornithine decarboxylase differentiates *P. penneri* from *P. mirabilis*.

TREATMENT

Proteus Infections

Intrinsic resistance occurs in all *Proteus* spp. to nitrofurantoin, polymyxins, imipenem, and the tetracycline derivatives (e.g., tigecycline, eravacycline, omadacycline) and, in *P. vulgaris* and *P. penneri*, also to ampicillin and the first- and second-generation cephalosporins. Acquired resistance (% of isolates) occurs in *P. mirabilis* to ampicillin (15–65%), and in *Proteus* spp. to fluoroquinolones (10–55%), fosfomycin (7–22%), and TMP-SMX (20–50%). In *P. mirabilis*, ampicillin-sulbactam is more active than ampicillin, with resistance prevalences of 6–18%, but the prevalence of ESBL production (which confers ampicillin-sulbactam resistance) is increasing in the United States (5–10%) and Asia (up to 60%). Isolates of *P. mirabilis* that produce CTX-M ESBLs have been recovered from ambulatory patients with no recent health care contact (see the section on the treatment of extraintestinal *E. coli* infections for treatment considerations). The use of third-generation cephalosporins can induce or select for stable de-repression of AmpC β-lactamase in *P. vulgaris*. Acquired carbapenem resistance remains relatively infrequent (<10%). However, production of a class B metallo-β-lactamase (e.g., NDM) limits treatment options due to the inherent resistance of *Proteus* spp. to polymyxins and tetracycline derivatives (see "Carbapenemase" above). For critically ill patients, agents with excellent activity overall against *Proteus* spp. (90–100% of isolates susceptible) include carbapenems (excepting imipenem), amikacin, piperacillin-tazobactam, aztreonam, cefepime, ceftazidime-avibactam, ceftolozane-tazobactam, and meropenem-vaborbactam.

ENTEROBACTER AND *CRONOBACTER* INFECTIONS

The *E. cloacae* complex is responsible for most *Enterobacter* infections, whereas *Cronobacter sakazakii* (formerly *Enterobacter sakazakii*), *Cronobacter malonaticus*, *Enterobacter cancergenus*, and *Enterobacter gergoviae* are less commonly isolated (<1% for each). *Enterobacter bugandensis* has been recently described as an agent of sepsis in neonates and was isolated from the International Space Station. *Enterobacter* spp. cause primarily health care–related infections. The organisms are widely prevalent in foods, environmental sources (including equipment at health care facilities), and a variety of animals.

Colonization with these organisms is uncommon among healthy humans, but increases significantly with LTCF residence or hospitalization. Although colonization is an important prelude to infection, direct introduction via IV lines (e.g., contaminated IV fluids or pressure monitors) or contaminated non-FDA-approved stem cell products also occurs. Extensive antibiotic resistance has developed in *Enterobacter* spp. and probably has contributed to these organisms' emergence as prominent nosocomial pathogens. Risk factors for *Enterobacter* infection include prior antibiotic treatment, comorbid disease, and ICU residency. *Enterobacter* spp. causes a spectrum of extraintestinal infections similar to those described for other GNB.

INFECTIOUS SYNDROMES

The most commonly encountered syndromes include pneumonia, UTI (particularly catheter-associated), intravascular device–related infection, surgical site infection, and abdominal infection (primarily postoperative or related to devices such as biliary stents). Nosocomial sinusitis, meningitis related to neurosurgical procedures (including use of intracranial pressure monitors), osteomyelitis, and endophthalmitis after eye surgery are less frequent. Neonates (particularly if low-birth-weight) are at risk for *C. sakazakii* infection, including neonatal bacteremia, necrotizing enterocolitis, and meningitis (which is often complicated by brain abscess or ventriculitis). Contaminated powdered infant formula has been implicated as a source for such neonatal infections. The WHO recommends that, to reduce the initial number of bacteria, powdered infant formula should be reconstituted with hot water (>70°C) and, to limit replication of residual bacteria, the reconstituted formula should be stored at <5°C or its storage time minimized.

Enterobacter bacteremia can result from primary infection at any anatomic site. In bacteremia of unclear origin, particularly in an outbreak setting, sources for consideration should include contaminated IV fluids or medications, blood components or plasma derivatives, catheter-flushing fluids, pressure monitors, and dialysis equipment. *Enterobacter* can also cause bacteremia in neutropenic patients. *Enterobacter* endocarditis is rare, occurring primarily in association with illicit IV drug use or prosthetic valves.

DIAGNOSIS

Enterobacter is readily isolated and identified in the laboratory. Most strains are lactose-positive and indole-negative.

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TREATMENT

Enterobacter Infections

E. cloacae is intrinsically resistant to ampicillin, ampicillin-sulbactam, ampicillin-clavulanate, the first-generation cephalosporins, and the cephamycins. The prevalence of acquired resistance has ranged from 15 to 40% for piperacillin-tazobactam, 5 to 23% for polymyxin E, 15 to 17% for fosfomycin, 15 to 30% for TMP-SMX, and 5 to 20% for fluoroquinolones and is ~10% for omadacycline (53% if tetracycline resistant). USNHSN data from 2015–2017 identified 8.9% of *E. cloacae* isolates as presumptively ESBL-producing, based on cefepime resistance. The prevalence of ESBLs in *E. cloacae* outside of the United States is 20–50%. The use of third-generation cephalosporins can induce or select for stable de-repression of AmpC β-lactamase. Because resistance may emerge during therapy (in one study, this phenomenon was documented in 20% of clinical isolates), these agents should be avoided in the treatment of serious *Enterobacter* infection.

Cefepime is stable in the presence of AmpC β-lactamases; thus, it is a suitable option for treatment of *Enterobacter* infections so long as no coexistent ESBL is present. Overall, resistance prevalence generally ranges from 10 to 25% for cefepime and 25 to 50% for aztreonam and the third-generation cephalosporins. Carbapenem resistance remains relatively uncommon (USNHSN data from 2015–2017 identified a 5% prevalence) and is more commonly associated with increased AmpC expression and decreased permeability due to porin mutations rather than carbapenemase production, although acquisition of carbapenemase genes is increasing (see "Carbapenemase" above). Uncertainty exists on the optimal treatment for non-CP-CR-*Enterobacter* spp. Fortunately, overall, the percentage of susceptibility is high (90–99%) for carbapenems, amikacin, plazomicin, ceftazidime-avibactam, meropenem-vaborbactam, imipenem/cilastatin-relebactam, cefiderocol, tigecycline, eravacycline, and omadacycline (the latter three for tetracycline-susceptible isolates). Once susceptibility data for a patient's isolate become available, de-escalation of the antimicrobial regimen is advisable whenever possible.

SERRATIA INFECTIONS

S. marcescens causes >90%, and *Serratia liquefaciens* complex <10%, of *Serratia* infections. Serratiae are found primarily in the environment (including in health care institutions), particularly in moist settings. Serratiae have been isolated from a variety of animals, insects, and plants, but only infrequently from healthy humans. In LTCFs and hospitals, reservoirs for the organisms include the hands and fingernails of health care personnel, food, milk (on neonatal units), sinks, medical equipment or devices, IV solutions or parenteral medications (particularly those generated by compounding pharmacies), prefilled syringes and multiple-access medication vials (e.g., for heparin, propofol, saline), blood products (e.g., platelets), hand soaps and lotions, irrigation solutions, and even disinfectants such as chlorhexidine.

Infection results from either direct inoculation (e.g., via contaminated injected substances [IV fluids, medications, or recreational drugs] or snake bite) or colonization (primarily of the respiratory tract). Sporadic infection is most common, but outbreaks (often involving MDR strains in adult and neonatal ICUs) also occur. Hygiene, medication-compounding standards, sterile technique, and infection control programs are critical measures to prevent infection.

The spectrum of extraintestinal infections caused by *Serratia* is similar to that for other GNB. *Serratia* species are usually considered to cause mainly health care–associated infections; they account for 1–3% of hospital-acquired infections. However, population-based laboratory surveillance studies in Canada and Australia have demonstrated that community-acquired *Serratia* infections occur more commonly than was previously appreciated, and case reports have documented serious infection in otherwise healthy hosts. *Serratia* also is one of the pathogens associated with chronic granulomatous disease.

INFECTIOUS SYNDROMES

The most common primary sites of *Serratia* infection are the respiratory and genitourinary tracts, intravascular devices, the eye (contact lens–associated keratitis and other ocular infections), surgical wounds, and the bloodstream (from contaminated infusions), although most episodes of *Serratia* bacteremia arise from one of the listed focal infections rather than contaminated infusate. Less common syndromes are soft tissue infections (including myositis, fasciitis, mastitis), osteomyelitis, abdominal and biliary tract infections (usually postprocedural), and septic arthritis (primarily from intraarticular injections). Serratiae are uncommon causes of neonatal meningitis; postsurgical meningitis, endophthalmitis, or breast implant infection; and bacteremia in neutropenic patients. Endocarditis is rare, occurring most commonly in IV drug users.

DIAGNOSIS

Serratiae are readily cultured and identified by the laboratory and are usually lactose- and indole-negative. The red pigmentation of some *S. marcescens* strains and *Serratia rubidaea* can produce distinctive clinical findings (e.g., pink breast milk or hypopyon; pseudohemoptysis).

TREATMENT

Serratia Infections

Most *Serratia* strains (>80%) are intrinsically resistant to ampicillin, amoxicillin-clavulanate, ampicillin-sulbactam, first- and second-generation cephalosporins, cephamycins, nitrofurantoin, and polymyxins; likewise, tetracycline derivatives are poorly active. By contrast, fluoroquinolones, TMP-SMX, piperacillin-tazobactam, fosfomycin, and omadacycline are active against 85–95% of U.S. and European isolates, including those resistant to tetracycline. Both in the United States and globally, the prevalence of ESBL-producing isolates is generally low (<10%), but rates of 20–30% have been reported in Asia and Latin America. The use of third-generation cephalosporins may result in the induction or selection of variants with stable de-repression of chromosomal AmpC β-lactamases during therapy but is uncommon. Resistance prevalence generally ranges from 10 to 20% for aztreonam and the third-generation cephalosporins. Acquisition of carbapenemase-encoding genes is uncommon but increasing. Production of a class B metallo-β-lactamase (e.g., NDM) limits treatment options due to *Serratia*'s predictable resistance to polymyxins and tetracycline derivatives (see "Carbapenemase" above). For critically ill patients, the most active agents overall (>90% susceptible) are carbapenems, piperacillin-tazobactam, cefepime, amikacin, plazomicin, ceftazidime-avibactam, ceftolozane-tazobactam, and meropenem-vaborbactam.

CITROBACTER INFECTIONS

C. freundii and *Citrobacter koseri* cause most human *Citrobacter* infections, which are epidemiologically and clinically similar to *Enterobacter* infections. *Citrobacter* species are commonly present in water, food, soil, and certain animals. Colonization with these organisms is uncommon among healthy humans, but increases significantly with LTCF residence or hospitalization. *Citrobacter* species account for 1–2% of nosocomial infections. The affected hosts are usually immunocompromised and/or have comorbid disease or disruption of skin or mucosal barriers. Infection from treatment with contaminated, non-FDA-approved stem cell products has been described. *Citrobacter* causes extraintestinal infections similar to those described for other GNB.

INFECTIOUS SYNDROMES

The urinary tract accounts for 40–50% of *Citrobacter* infections. Less commonly involved sites include the biliary tree (particularly with stones or obstruction), the respiratory tract, surgical sites, soft tissue (e.g., decubitus ulcers), the peritoneum, and intravascular devices. Osteomyelitis (usually from a contiguous focus), central nervous system infection in adults (from neurosurgical or other types of meningeal

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disruption), and myositis occur rarely. *Citrobacter* (primarily *C. koseri*) also causes 1–2% of neonatal meningitis cases, of which 50–80% are complicated by brain abscess. Further, case reports in adults suggest that *C. koseri* infection has a predilection for abscess formation. *Citrobacter* bacteremia is most often due to UTI, biliary/abdominal infection, or intravascular device infection, and occurs in some neutropenic patients. Endocarditis and other endovascular infections are rare.

DIAGNOSIS

Citrobacter species are readily isolated and identified; 35–50% of isolates are lactose-positive, and 100% are oxidase-negative. *C. freundii* is indole-negative, whereas *C. koseri* is indole-positive.

TREATMENT

Citrobacter Infections

C. freundii is more extensively antibiotic-resistant than is *C. koseri*. Most *C. freundii* isolates are intrinsically resistant to ampicillin, ampicillin-sulbactam, amoxicillin-clavulanate, first-generation cephalosporins, and cephamycins. *C. koseri* exhibits intrinsic resistance to ampicillin and ampicillin-sulbactam. Overall, the prevalence of acquired resistance generally ranges from 15 to 35% for third-generation cephalosporins, piperacillin-tazobactam, fluoroquinolones, and TMP-SMX and is ~10% for nitrofurantoin and omadacycline (but 39% for omadacycline if tetracycline-resistant). The prevalence of ESBL production ranges from 5 to 30%. The use of third-generation cephalosporins may result in the induction or selection of variants with stable de-repression of chromosomal AmpC β-lactamases during therapy. Presently, <10% of isolates have acquired carbapenemases (see "Carbapenemase" above). Carbapenems, amikacin, plazomicin, fosfomycin, polymyxins, cefepime, ceftazidime-avibactam, cefiderocol, tigecycline, eravacycline, and omadacycline (the latter three if tetracycline-susceptible) are the most active agents against *Citrobacter* isolates (>90% susceptible).

MORGANELLA AND *PROVIDENCIA* INFECTIONS

M. morganii, Providencia stuartii, and (less frequently) *Providencia rettgeri* are the members of their respective genera that cause systemic human infections. *P. alcalifaciens* has been implicated as a cause of food-borne gastroenteritis. These organisms' epidemiologic associations, pathogenic properties, and clinical manifestations resemble those of *Proteus* species. *Morganella* and *Providencia* occur more commonly among LTCF residents than among hospitalized patients, largely resulting from chronic urinary catheter use. Because of these organisms' intrinsic resistance to polymyxins and tigecycline, they may become increasingly common in settings with extensive use of these agents.

INFECTIOUS SYNDROMES

These species are primarily urinary tract pathogens, causing UTIs that are most often associated with long-term (>30-day) catheterization. Such infections commonly lead to biofilm formation and catheter encrustation (sometimes causing catheter obstruction) or the development of struvite bladder or renal stones (sometimes causing renal obstruction, abscess, and extrarenal extension, and serving as foci for relapse). They can cause purple urine ("purple bag syndrome"), as can *P. mirabilis*, *K. pneumoniae*, *E. coli*, and *P. aeruginosa*. *Morganella* is also commonly isolated from snakebite infection.

Other, less common infectious syndromes due to *Morganella* and *Providencia* include surgical site infection, soft tissue infection (primarily involving decubitus and diabetic ulcers), burn site infection, pneumonia (particularly ventilator-associated), intravascular device infection, and intraabdominal infection. Rarely, the other extraintestinal infections described for GNB also occur. Bacteremia is uncommon; when it does occur, any infected site can serve as the source, but the urinary tract accounts for most cases, followed by surgical site, soft tissue, and hepatobiliary infections.

DIAGNOSIS

M. morganii and *Providencia* are readily isolated and identified. Nearly all isolates are lactose-negative and indole-positive.

TREATMENT

Morganella and *Providencia* Infections

Morganella and *Providencia* are intrinsically resistant to ampicillin, ampicillin-clavulanate, ampicillin-sulbactam, first-generation cephalosporins, nitrofurantoin, tetracyclines and derivatives (e.g., tigecycline), imipenem (but not the other carbapenems), and the polymyxins. *P. stuartii* additionally exhibits intrinsic resistance to gentamicin and tobramycin, as does *M. morganii* to second-generation cephalosporins. Fosfomycin is poorly active (>50% resistance). The prevalence of resistance generally ranges from 10 to 30% for the third-generation cephalosporins, from 10 to 40% for fluoroquinolones, and from 20 to 40% for TMP-SMX; the prevalence is more widely variable for piperacillin-tazobactam. The prevalence of ESBL production is generally <10%. The use of third-generation cephalosporins can induce or select for stable de-repression of AmpC β-lactamase for both *Morganella* and *Providencia*. The prevalence of acquired carbapenemase production is <10%. Production of a class B metallo-β-lactamase (e.g., NDM) limits treatment options due to the inherent resistance of the *Proteeae* to polymyxins and tetracycline derivatives (see "Carbapenemase" above). Overall, the most active agents (>90% of isolates susceptible) are carbapenems (excepting imipenem), amikacin, cefepime, ceftazidime-avibactam, ceftolozane-tazobactam, meropenem-vaborbactam, and cefiderocol. Removal of a colonized urinary catheter or stone is critical for eradication of UTI.

EDWARDSIELLA INFECTIONS

E. tarda is the only member of the genus *Edwardsiella* that is associated with human disease. This organism is found predominantly in freshwater and marine environments and in the associated aquatic animal species. Human acquisition occurs primarily from interaction with these reservoirs or ingestion of raw or inadequately cooked aquatic animals. *E. tarda* infection is rare in the United States, where acquisition occurs mainly along the Gulf of Mexico; recently reported cases are mostly from Asia. This pathogen shares clinical features with *Salmonella* species (as an intestinal pathogen; **Chap. 165**), *Vibrio vulnificus* (as an extraintestinal pathogen; **Chap. 168**), and *Aeromonas hydrophila* (as both an intestinal and an extraintestinal pathogen; **Chap. 158**).

INFECTIOUS SYNDROMES

Gastroenteritis is the predominant *Edwardsiella*-associated infectious syndrome (50–80% of reported cases). Self-limiting watery diarrhea is most common, but severe colitis also occurs. The most common extraintestinal infection is wound infection due to direct inoculation, which is often associated with brackish or freshwater injuries, snakebites, or fish-related trauma. A case of pneumonia occurred after a near-drowning incident. Cholecystitis, cholangitis, and hepatic abscess may be due to ascending infection via the biliary tree. Other infectious syndromes result from invasion of the gastrointestinal tract and subsequent bacteremia. A primary bacteremic syndrome, sometimes complicated by meningitis, has a 40% case–fatality rate; hematogenous seeding may result in hepatic and intra- and extraperitoneal abscesses, endocarditis, mycotic aneurysm, septic arthritis, osteomyelitis, necrotizing fasciitis, and empyema. Most hosts who develop systemic *Edwardsiella* infection have significant comorbidities (e.g., hepatobiliary disease, iron overload, cancer, or diabetes mellitus).

DIAGNOSIS

Although *E. tarda* can readily be isolated and identified, most laboratories do not routinely screen for or identify it in stool samples. Production of hydrogen sulfide is a characteristic biochemical property.

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TREATMENT

Edwardsiella Infections

E. tarda is susceptible to most antimicrobial agents appropriate for use against GNB. Gastroenteritis is generally self-limiting, but treatment with a fluoroquinolone may hasten resolution. In the setting of severe sepsis, fluoroquinolones, third- and fourth-generation cephalosporins, carbapenems, and amikacin—either alone or in combination—are the safest choices pending susceptibility data.

INFECTIONS CAUSED BY MISCELLANEOUS GENERA

Other gram-negative organisms such as *Hafnia*, *Kluyvera*, *Cedecea*, *Pantoea*, *Ewingella*, *Leclercia*, *Raoultella*, and *Photorhabdus* spp. are occasionally isolated from diverse clinical specimens, including blood, sputum, urine, cerebrospinal fluid, joint fluid, bile, and wounds. Such organisms cause infection predominantly in compromised hosts or in association with an invasive procedure or foreign body. Cephalosporinases from *Kluyvera* have been implicated as the progenitors of CTX-M ESBLs. *Kluyvera* and *Raoultella* may produce carbapenemases.

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162 *Acinetobacter* Infections

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DEFINITION

Acinetobacter species were first described in 1911 and named *Micrococcus calcoaceticus*. Thereafter, the genus was renamed multiple times; since 1950, it has been known as *Acinetobacter*. *Acinetobacter* species are gram-negative, oxidase-negative, nonmotile, nonfermenting coccobacilli that are easily recovered on standard culture media. Differentiation among *Acinetobacter* species on the basis of phenotypic characteristics alone is very difficult. Molecular-based methods such as matrix-assisted laser desorption–ionization–time-of-flight mass spectrometry (MALDI-TOF-MS) and quantitative real-time polymerase chain reaction (PCR) are usually necessary to identify *Acinetobacter baumannii*, the most clinically relevant species of the genus.

ETIOLOGY AND EPIDEMIOLOGY

Acinetobacter species are naturally encountered in water and soil and have also been recovered from fruits and vegetables. In humans, *Acinetobacter* can be found on the skin and in the respiratory and gastrointestinal tracts. *A. baumannii* is capable of surviving environmental desiccation for weeks; this characteristic is important from an infection-control perspective as it allows this organism to persist in the hospital environment and on equipment.

Acinetobacter was historically considered a pathogen of hot and humid climates. In recent years, however, hospital outbreaks caused by *A. baumannii* have been reported worldwide, even in temperate climates. In the United States, the Centers for Disease Control and Prevention (CDC) estimates that 12,000 *Acinetobacter* infections occur every year, 7300 of which are caused by multidrug-resistant strains, with 500 attributable deaths. The increase in the number of infections with *A. baumannii* is suspected to be due to the rapid spread of certain genetically distinct lineages; of the three international clonal lineages (ICLs), ICL I and ICL II are multidrug resistant. The predominance of these lineages remains unexplained, although it has been proposed that this population structure is the result of two waves of expansion. The first wave followed a bottleneck (possibly linked to a restricted ecologic niche) that occurred in the distant past. The second wave is ongoing and is being driven by the rapid expansion of a limited number of multidrug-resistant clones.



Analysis of the *A. baumannii* pangenome (the sum of the core and dispensable genomes) has shown that its organization is characterized by a small core genome and a large accessory or disposable genome. This organization reflects *A. baumannii's* high plasticity, which enables it to acquire exogenous genetic material. With few exceptions, gene functions associated with virulence are found in the core genome; this observation suggests a limited role for the acquisition of new virulence traits in the recent nosocomial expansion of *A. baumannii* clones. Genes associated with resistance to antimicrobial agents are found in both the species core genome and the accessory genome. In the accessory genome, these genes have been found in alien islands, often flanked by integrases, transposases, or insertion sequences. This pattern suggests possible acquisition by horizontal gene transfer from other *Acinetobacter* strains or even from different bacterial species present in the immediate environment. Acquisition of these antimicrobial resistance genes is hypothesized to have led to the recent rapid expansion of highly homogeneous clonal lineages, whose main difference from nonclonal *A. baumannii* appears to be their antimicrobial resistance.

Health Care–Associated Infections Infections caused by *A. baumannii* occur frequently among patients admitted to intensive care units (ICUs). Risk factors for colonization and infection with this pathogen include nursing home residence, prolonged ICU stay, central venous catheterization, tracheostomy, mechanical ventilation, enteral feedings, and treatment with third-generation cephalosporins,